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Improving automated extinction learning through robust off-policy reinforcement learning

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Abstract

Hintergrund: Angststörungen sind weltweit weit verbreitet. Expositionstherapie ist eine Behandlungsmethode der ersten Wahl, deren Verfügbarkeit jedoch aufgrund hoher Kosten und eines Mangels an Therapeut:innen eingeschränkt ist. Die Lücke zwischen dem Bedarf und der Verfügbarkeit professioneller Behandlung könnte durch eine Automatisierung der Expositionstherapie verringert werden, wodurch gleichzeitig deren Zugänglichkeit, Kosteneffizienz und Wirksamkeit verbessert werden könnten. Insbesondere Off-Policy Reinforcement Learning (RL) Algorithmen könnten knappe Patientendaten effizient nutzen, um eine robuste Expositionstherapie zu ermöglichen.

Methoden: In einer simulierten Umgebung wurde ein Off-Policy RL Agent darauf trainiert, Extinktionslernen (EL), ein Labormodell der Expositionstherapie, an drei Patientenmodellen durchzuführen. In vier Experimenten wurden die Auswirkungen der Zusammensetzung und Menge vergangener simulierter Daten auf das erfolgreiche Erlernen von EL untersucht. Mithilfe einer Simulation-zu-Simulation (Sim2Sim) Evaluation wurde außerdem untersucht, ob das erlernte EL robust auf andere Patientenmodelle und Patientenparameter generalisiert, die während des Trainings nicht verwendet wurden.

Ergebnisse: Der RL Agent erlernte erfolgreich die Durchführung von EL in der simulierten Umgebung und wendete dabei am häufigsten die unmittelbare Extinktion an, eine gängige Strategie des EL. Durch die Verwendung von Replay Buffern, die mit zufälligen Aktionen vorbefüllt wurden, beschleunigte der Off-Policy DQN Algorithmus das Erlernen sinnvoller EL-Strategien im Vergleich zum Lernbeginn mit leeren Buffern. Der Agent führte das Extinktionslernen robust über verschiedene Patientenmodelle hinweg durch und zeigte, mit einigen Einschränkungen, auch innerhalb desselben Patientenmodells robuste Leistungen bei unterschiedlichen Parameterkonfigurationen.

Schlussfolgerung: Der RL-Agent zeigte eine effektive und generalisierbare Leistung innerhalb der untersuchten simulierten Extinktionslernumgebung. Die Ergebnisse sind jedoch durch die Einfachheit und Ähnlichkeit der Simulationen, den eindimensionalen

Beobachtungsraum, den eingeschränkten Aktionsraum sowie die nur teilweise angewandte Randomisierung der Parameter innerhalb eines Modells begrenzt. Darüber hinaus wurde eine in der Literatur empfohlene Benchmark Evaluation nicht durchgeführt.

Keywords: Reinforcement Learning, Mental Health, Exposure Therapy, Extinction Learning

Improving automated extinction learning through robust off-policy reinforcement learning

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1 Introduction

Phobia and anxiety disorder have a strong impact on mental health and overall quality of life worldwide. While effective treatments exist, a gap between the demand and the availability of professional anxiety disorder treatments is prevalent (Alonso et al., 2018). 9.8 % of respondents from 21 countries were diagnosed to suffer from anxiety disorders (Alonso et al., 2018) within a 12-months period. Although 41.3% of the diagnosed perceived a need for professional treatment, only 9.8% of all respondents actually received adequate treatment. Another study estimated the global prevalence of anxiety disorders with 4.05% and observed a sharp increase of 55% between 1990 and 2019 (Javaid et al., 2023). With anxiety prevalence continuing to increase and a substantial gap between the perceived need for treatment and the actual receipt of treatment (Alonso et al., 2018), negative consequences for affected individuals and society as a whole are expected.

The first line effective treatment against anxiety disorders is exposure therapy (ET) (Abramowitz et al., 2019; Telch et al., 2014). In ET, a therapist helps patients confront their fears by repeatedly exposing them to stimuli or situations that trigger anxiety. Throughout the process, the therapist aims to change how patients respond to these triggers and to improve their ability to cope with anxiety. Typically, ET is tailored to each patient’s individual needs and clinical history. ET is conducted over multiple sessions depending on the treatment plan and the patient’s progress. In addition to anxiety disorders, ET is widely used in the treatment of panic (Meuret et al., 2012), stress (Rothbaum & Schwartz, 2002), and obsessive-compulsive disorders (Foa, 2010).

Underlying ET are the paradigms of associative and extinction learning (EL) (Craske et al., 2014). *Associative learning* is the process of acquiring an association between an unconditioned stimulus (US), which elicits a naturally occurring behavior or unconditioned response (UR), and a neutral stimulus (NS), which would have not produced any response. After an association is formed, the NS leads to a similar outcome as the US. Now, the NS is called a conditioned stimulus (CS), which leads to a conditioned response

(CR). *Extinction learning* postulates that any learned association can be unlearned by repeatedly observing the CS in the absence of the US. Unlearning the association extinguishes or lowers the probability of the CR occurrence. To study ET and its application in anxiety disorders it is common to use associative learning as a simplified laboratory (lab) model for fear acquisition and EL as a simplified lab model for ET (Craske et al., 2008, 2014).

The gap between the demand for anxiety disorder treatments and the availability of professional ET might be reduced by applying artificial intelligence (AI) systems to automate or assist ET. One promising approach within this broader effort is reinforcement learning (RL), a class of AI methods that model interactions over time between a reward-maximizing agent and an environment (Sutton & Barto, 2018). In RL, the environment provides the agent with a reward signal at each time step, encoding how desirable the current outcome is with respect to the goal. The agent’s objective is to maximize its expected cumulative reward by selecting, at each time step, an action from a defined set that causes the environment to transition from the current state to another subsequent state. Depending on the state, transition dynamics, and action, the agent obtains rewards and adapts its policy to maximize its expected cumulative reward. A policy is a mapping from different states to actions. In other words, RL seeks to find for each state an optimal action to maximize its expected cumulative reward over the course of interaction. Training an RL agent refers to its repeated interaction with the environment and the subsequent updates of its policy, whereas testing refers to evaluating the learned policy in terms of achieved rewards or the quality of the learned state-action mappings. To date, RL has been applied to improve medical diagnosis, process control in surgical procedures, drug discovery, health management, and the design of dynamic treatment plans, among other areas. In their survey, Yu et al. (2021) offer a comprehensive overview of how RL has been used to facilitate or enhance various healthcare tasks.

In an ET setting, the RL agent could correspond to an (RL) therapist or a

decision-support system, while the environment could comprise the patient suffering from an anxiety disorder and all relevant clinical factors that evolve over the course of ET. In this analogy, states can correspond to the patient’s current condition, approximated by self-reported anxiety levels, psychophysiological indicators (e.g., skin conductance, heart rate), or contextual features of the exposure situation. Actions can correspond to therapeutic decisions, such as which stimulus to present (e.g., image of a spider, or an instruction to think of a spider), for how long, and with what contingency relative to other events. With the overall goal of reducing or extinguishing the patient’s anxiety, each therapeutic action influences the patient’s transition into subsequent anxiety states that are uncertain because they depend on both the chosen intervention and the patient’s individual learning dynamics. For each interaction the therapist needs to find the optimal therapeutic action to reduce the patients anxiety levels. Thus, the reward could be any signal of therapeutic progress (e.g., patient feedback, measure of anxiety-level reduction) and the policy could correspond to a treatment protocol and provide the therapists with a mapping from the anxiety levels of the patient to therapeutic actions required.

Conceptually, RL is well suited for automating or assisting therapists conducting ET, especially when using the simplified lab model of EL. EL reduces ET to a tractable learning and unlearning task in which a CR, that is anxiety, is acquired and then reduced through repeated exposure to the CS. This simplification makes it possible to formalize ET as a sequential decision-problem with defined states, actions, and rewards, while still capturing core mechanisms relevant to clinical ET. In this setting, RL could help to find the best actions for a given state to reduce the patients anxiety. Moreover, through its policy RL could provide highly individual actions tailored to the needs of each patient and its current anxiety state, thus enabling sophisticated patient-tailored therapeutic interventions. At the same time, it acknowledges that EL does not account for all potentially complex cognitive, emotional, and social processes underlying mental disorders, and thus using EL performed by RL therapists serves as an initial exploration of developing and evaluating

RL-based therapeutic agents before extending them to more comprehensive models.

In this setting, the resulting system can be used in different ways. In a human-in-the-loop configuration, the presentation of the stimulus is carried out by a human therapist, with the RL agent providing complementary decision support (e.g., recommendations for the next exposure step). In a fully automated configuration, stimulus presentation is delivered digitally on a screen or through virtual reality (VR), and the RL agent directly controls the exposure treatment protocol. Previous research has already shown promising first attempts to automate EL (Mahmoudi-Nejad et al., 2024; Tomasiak, 2024). Tomasiak (2024) explored parameter configurations that allowed an RL agent to perform EL on a simulated patient model that had previously learned associations in a two-stimulus setting. A computational patient model was used instead of real patients and simulated their associative fear learning and extinction. Moreover, Mahmoudi-Nejad et al. (2024) used an RL agent to dynamically generate spider images that induced pre-defined target levels of fear in human participants.

Research Gap

Nevertheless, important research gaps remain on the path toward automated or RL-assisted ET. Firstly, Tomasiak (2024) demonstrated that RL agents can learn to perform EL on synthetic patient models within an associative-learning and EL paradigm. Yet, RL agents were trained to perform EL on simulated patient models without incorporating established EL protocols or treatment strategies. For instance, one standard EL protocol is immediate extinction, in which the CS is repeatedly presented in the absence of the US. Incorporating established EL protocols might enable the RL agent to learn more clinically meaningful policies. Data from past therapist-patient interaction might allow RL agents to learn from patient characteristics that have not been adequately represented in the current simulations, like comorbidities, extreme levels of (anxiety) disorders, or learning impairments. Moreover, training on such past data could be particularly valuable in domains where data are scarce or costly to obtain, such as in

personalized therapeutic interventions. Hence, it remains unclear how RL agents perform when trained not merely on patient model simulations, but also on data incorporating established EL protocols or past therapist-patient interactions into training the RL agent, and whether this would improve EL performance or generalization to previously unseen therapeutic contexts (e.g., patient properties not reflected in the simulation).

Secondly, it remains unclear whether RL models can robustly learn to extinguish learned associations in contexts that differ from those used during training or the development of the RL therapist. To date, no simulation-to-simulation (Sim2Sim or model mismatch robustness) research has examined whether an RL therapist trained on simulations derived from one patient model can reliably perform ET on other patient models simulations. In Tomasiak (2024), both training and evaluation of the RL agent were conducted on the same patient models. In Mahmoudi-Nejad et al. (2024) neither EL nor Sim2Sim was applied. However, robust Sim2Sim performance is essential to determine whether reported EL outcomes are limited to specific models and settings or can be generalized to patient models not used during training. Beyond simulations, the present thesis hypothesizes that strong Sim2Sim performance may be a necessary precondition for reliable real-world application with patients who have diverse characteristics and across varied contexts, although this remains to be empirically validated. Given the potential risks that RL therapists might pose to human study participants and patients, strong evidence of their generalizability across different contexts is required before RL therapists can be tested or applied in studies involving real patients.

Contribution to Research

This thesis sets out to close these research gaps on the way to automated or RL-assisted ET. It contributes to the literature in two ways. Firstly, it examines whether an RL therapist is capable of learning to perform EL tailored to the individual states of a patient model by using past patient model interaction data. Hence, it explores the possibility of RL assisting or complementing human therapists and performs preliminary

simulation studies to evaluate its performance. Moreover, it explores whether incorporating past data from established ET protocols into training can increase the rewards achieved by RL agents, indicating more effective EL performance. To allow learning from past data the off-policy RL algorithm can be used. Off-policy learning allows the RL agent to learn from policies that deviate from the one it is using itself. The past training data could stem from previous EL protocols performed on simulated patients. In addition this thesis estimated the least required amount of patient data for the RL therapist to successfully learn performing ET. Hence, this thesis constitutes a preliminary simulation study before collecting real world patient data to optimize ET performance and is an important step to improve and apply RL enabled ET.

Secondly, this thesis conducts Sim2Sim experiments to evaluate the robustness of the EL performance. Three different computational models of human associative learning were used to simulate patient behavior: the Rescorla-Wagner (RW) (Rescorla & Wagner, 1972), the Kalman-filter-RW (KRW) (Gershman, 2015), and the Dirichlet-process-KRW (DPKRW) (Gershman et al., 2017). In Sim2Sim the training is conducted on one patient model, while its performance is tested on another. This approach allows for the assessment of the therapist’s adaptability across diverse patient parameters and models. A robust performance might indicate low over-fitting to one particular patient model and demonstrate good performance on the other patient models investigated. Each model in use, has its own advantages and limitations in simulating empirical human decisions and behaviors. While the latter models, KRW and DPKRW, extend and generalize the RW model to resolve empirical inconsistencies, all three are utilized in this thesis to enhance comparability with prior research. Moreover, in this thesis a robust performance is hypothesized to indicate a robust performance on other models not used in this thesis, future model extensions and generalizations, or real world patients with differing characteristics, although this remains to be evaluated in future experiments.

1.1 Research Questions

The goal of this thesis is to advance the research on the path to automated or RL-assisted ET. Thus, the thesis will address the two identified research gaps with the following two research questions (Q), and their sub-questions (SQ).

1.1.1 *Training RL Agents on Past Data from Simulated EL Sessions*

Q1 To what extent can an off-policy RL algorithm learn an effective EL policy from simulated interaction data generated by RL agents and patient models?

SQ1.1 What effect does the composition of the past data have on EL performance by RL therapists, and what specific strategies do RL therapists learn from different past data compositions?

SQ1.2 What effect does the amount of past data have on EL performance by the RL therapist?

1.1.2 *Robustness of EL Performance Across a Set of Patient Models*

Q2 How robustly do off-policy RL therapists perform EL?

SQ2.1 How well do off-policy RL agents perform EL on patient models that differ from the one used during training?

SQ2.2 How well do RL therapists maintain EL performance across different patient model parameter specifications that differ from those used during training?

2 Theoretical Background

2.1 Modeling Human Associative Learning and Behavior

Understanding human learning and behavior is key in predicting, explaining, and altering the behavior of patients. In the early 1900s, Pavlov conducted his experimental research on animal learning (Pavlov, 1927) and introduced the framework of classical conditioning (CC). Pavlov observed that dogs produced saliva as a biologically predetermined response after seeing or anticipating food. Pavlov then repeatedly presented the food concurrently with a bell tone. Subsequently, the dog started to produce saliva upon hearing the bell tone, even in the absence of food, thus learning the association between the bell tone and the anticipation of receiving food. Quickly, CC experiments became the standard procedure to study the formation of associations between stimuli (or cues) with their outcomes. In CC, a stimulus that naturally elicits any behavioral response is called an unconditioned stimulus (US) and its response unconditioned response (UR). Learning is the acquisition of an association between a US with a neutral stimuli (NS), which previously did not lead to any UR. After the acquisition, the NS becomes a conditioned stimulus (CS) or cue and leads to a conditioned response (CR), which resembles the UR caused by the US.

Additionally, Pavlov observed that behaviors learned within an acquisition phase can be unlearned in an extinction phase. After several trials of presenting the bell tone without food, the dog produced progressively less saliva and eventually ceased its production entirely upon hearing a bell tone. Hence, the dog unlearned the previously acquired association. In CC an UR is elicited by a US, naturally due to biological dispositions. In contrast, a CR is a learned behavior occurring as a result of the repeated observation of an NS paired with a US, and in turn it can be unlearned. Extinguishing a CR can be achieved by the repeated observation of the CS, when it is not paired with the US anymore. Each extinguishing trial tends to lead to a lower intensity of the CR, until it eventually vanishes.

Application of the CC framework in the field of psychology proved to be a powerful

method to study human behavior. CC experiments with humans were widely replicated and resulted in similar learning patterns as observed in mammals (e.g., Lipp et al., 2003; Morís et al., 2017; Nadler et al., 2011; Shiban et al., 2015; Talmi et al., 2008; van den Akker et al., 2015). The psychological paradigm of behaviorism aims to study and explain observable behavior without relying on the (self-)reporting of subjective inner processes. Here, the elegance of the CC framework became evident. It enabled researchers to study, reproduce, and variate human learning behaviors in a randomized and highly controlled experimental setting without relying on reported measures. Most importantly for researchers, CC posits clear predictable, testable behaviors and (association) learning patterns. Expanding on the idea of predictability, Rescorla and Wagner (1972) introduced a mathematical formalization of human learning, the RW model.

RW Patient Model. RW quantifies learning as the change of the associative strength between a stimulus and its outcome at a given time point or trial. In other words, it estimates the learner’s expectation of a certain outcome given that a stimulus has been shown. Each acquisition (extinction) trial tends to increase (decrease) the associative strength within the observer. The RW model approximates the change of the associative strength on a per trial level and can be viewed as a point estimator (Gershman, 2015) of the learned association. Furthermore, psychophysiological changes of the CR (e.g., skin conductance, pupil size, heart period, respiration, and startle eye-blink) are real-world indicators of the change in associative strength (Bach & Melinscak, 2020). Hence, the learned association becomes testable after each trial. Moreover, the RW model allows to reflect differences in learning across humans. Human traits, which might affect learning, or other dispositional factors could be reflected in distinct model parameter values (Pike & Robinson, 2022). In the context of the EL experiments, the RW model predicts the patient model’s associative strength, which could be mapped to the CR level.

Mathematically RW can be formulated in the following equations (Gershman, 2015). In Equation 1, v_n describes the overall strength of association between the cues and

the US in trial n .

$$v_n = \mathbf{w}_n^T \cdot \mathbf{x}_n \quad (1)$$

In other words, it is the patient model's prediction of the occurrence of the US. On the right-hand side of the equation, $\mathbf{w}_n^T$ is a vector of the model's current weights for each stimulus and $\mathbf{x}_n$ is a binary vector indicating the presence of all stimuli on trial n (cue A or cue B). Learning is guided by a weight update rule given in Equation 2,

$$\mathbf{w}_{n+1} = \mathbf{w}_n + \alpha \cdot \mathbf{x}_n \cdot \lambda_n \quad (2)$$

where the updated weights for trial $n + 1$ are computed after learning in trial n . Here, α is the constant learning rate of the patient model. The learning rate determines the magnitude of weight updates after each trial. λ_n is the prediction error computed in Equation 3.

$$\lambda_n = US_n - (v_n) \quad (3)$$

KRW Patient Model. The KRW model (Gershman, 2015) is the Kalman-filter based Bayesian generalization of the RW model. Essentially, it describes learners as updating probabilistic beliefs about their associations, which are subject to uncertainty. In other words, learners have prior beliefs about their responses when observing a set of stimuli. With each trial, they collect new evidence that updates their prior beliefs into posteriors. With a growing number of trials, the uncertainty diminishes and the patients' beliefs become more accurate.

The KRW is modeled by a linear-Gaussian dynamical system (LDS), described in Equations 4-6. The learner's *prior* over weights $\mathbf{w}_0$ is a zero-mean Gaussian distribution, with covariance $\sigma_w^2 I$, expressing that initially there is no systematic association:

$$\mathbf{w}_0 \sim \mathcal{N}(0, \sigma_w^2 I) \quad (4)$$

The learners weights evolve over time as a random walk:

$$\mathbf{w}_n \sim \mathcal{N}(\mathbf{w}_{n-1}, \tau^2 I) \quad (5)$$

where constant τ^2 controls the volatility of the weight change. Likewise, the observed reward r_n is an Gaussian variable with mean equal to the model's expectation v_n of the US and variance σ_r^2 :

$$r_n \sim \mathcal{N}(v_n, \sigma_r^2) \quad (6)$$

Combining this LDS with Bayes' rule in Equation [7](#),

$$p(\mathbf{w}_n | \mathbf{x}_{1:n}) \propto p(\mathbf{x}_{1:n} | \mathbf{w}_n) p(\mathbf{w}_n) \quad (7)$$

the patient's posterior over association weights is Gaussian, with the mean updated according to Equation [8](#),

$$\hat{\mathbf{w}}_{n+1} = \hat{\mathbf{w}}_n + \mathbf{K}_n \lambda_n \quad (8)$$

where the Kalman gain $\mathbf{K}_n$ in Equation [10](#) is replacing the RW model's learning rate α , and covariance matrix Σ_n in Equation [9](#). The Kalman gain is stimulus-specific, and grows dynamically with higher uncertainty expressed in the diagonal of the covariance matrix.

$$\Sigma_{n+1} = \Sigma_n + \tau^2 \mathbf{I} - \mathbf{K}_n \mathbf{x}_n^T (\Sigma_n + \tau^2 \mathbf{I}), \quad (9)$$

$$\mathbf{K}_n = \frac{(\Sigma_n + \tau^2 \mathbf{I}) \mathbf{x}_n}{\mathbf{x}_n^T (\Sigma_n + \tau^2 \mathbf{I}) \mathbf{x}_n + \sigma_r^2} \quad (10)$$

DPKRW. The DPKRW model is based on Gershman et al. ([2017](#)). Extending KRW, DPKRW assumes that the learner infers a posterior probability distribution over latent causes that determine associations between stimuli and the US. This allows the model to store multiple associations, each linked to a different latent cause. This property is expressed by an additional dimension k for each latent cause, restricted to $k = 1, \dots, k_{max}$, so that each latent cause maintains its own association weights and covariance structure. For each latent cause k the weights are updated according to Equation [11](#):

$$\mathbf{w}_{k,n+1} = \mathbf{w}_{k,n} + \mathbf{K}_{k,n} \lambda_n \quad (11)$$

The prediction error λ_n is computed in Equation [12](#)

$$\lambda_n = r_n - (v_n^T \mathbf{p}_{Z_{n+1}}) \quad (12)$$

where $v_n = W_n x_n$ is the predicted US strength per latent cause k , obtained by multiplying the weights matrix with the presented stimuli, given the posterior probability $p_{Z_{n+1}}$ over all k_{max} latent causes. r_n is the binary variable indicating the presence of the US. Likewise, uncertainty is represented for each latent cause in a separate covariance matrix updated as in Equation 13

$$\Sigma_{k,n+1} = \Sigma_{k,n} + \tau^2 \mathbf{I} - p_{Z_{k,n}} (K_{k,n} (\mathbf{x}_n^T (\Sigma_{k,n} + \tau^2 \mathbf{I}))) \quad (13)$$

The Kalman gain $K_{k,n}$ is calculated in Equation 14, where $p_{Z_{k,n+1}}$ is the posterior probability for latent cause k :

$$K_{k,n} = \frac{p_{Z_{k,n+1}} (\Sigma_{k,n} + \tau^2 \mathbf{I}) \mathbf{x}_n}{p_{Z_{k,n}}^2 S_{k,n}} \quad (14)$$

with $S_{k,n}$ being the residual variance, calculated as in Equation 15.

$$S_{k,n} = \mathbf{x}_n^T (\Sigma_{k,n} + \tau^2 \mathbf{I}) \mathbf{x}_n + \sigma_r^2 \quad (15)$$

Additionally, the posterior distribution over latent causes is reflected in the term $p_{Z_{k,n}}$

$$p_{Z_{n+1}} = \frac{e^{\log(p_{Z_n}) + \ell_n}}{\sum e^{\log(p_{Z_n}) + \ell_n}} \quad (16)$$

with k_{max} -dimensional prior p_{Z_n} and the log-likelihood ℓ_n . Through the *Chinese restaurant process*, the prior is biased to stick to the old latent cause of the previous trial, but can produce unlimited new latent causes (Gershman et al., 2014). By default, the patient initially infers one latent cause. In Equation 17 a new latent cause is activated with probability determined by the concentration parameter α_{DP} , while the stickiness parameter β increases the probability of the previous latent cause. Here, g_n is the total number of the patient's so far activated latent causes, and h_n is the index of the patient's currently inferred latent cause

$$p_{Z_{k,n+1}} = \begin{cases} \alpha_{DP} & \text{if } k = g_n + 1 \rightarrow \text{activate new latent cause} \\ p_{Z_{k,n}} + \beta & \text{if } k = h_n \rightarrow \text{currently active latent cause} \\ p_{Z_{k,n}} & \text{otherwise} \end{cases} \quad (17)$$

Subsequently, the k_{max} -dimensional prior over all latent causes is normalized by $p_{Z_n} \frac{p_{Z_n}}{\sum p_{Z_n}}$ to form a probability distribution over latent causes. The log-likelihood ℓ_n is normally distributed, as in Equation 18

$$\ell_n = -\frac{1}{2} \left(\frac{r_n - W_n x_n}{\sqrt{S_n}} \right)^2 - \log \left(\sqrt{2\pi} \sqrt{S_n} \right) \quad (18)$$

DPKRW selects the latent cause L_{n+1} , which will be active in the next trial by the maximum a posteriori (MAP) of the posterior distribution.

$$L_{n+1} = \operatorname{argmax} p_{Z_{n+1}} \quad (19)$$

2.2 Exposure Therapy for Phobias and Anxiety Disorders

In the 1950's behavioral psychologists developed exposure therapy (ET) by applying findings from CC experiments. Today, ET is the first-line treatment for anxiety disorders such as specific phobias and obsessive-compulsive disorder (Abramowitz et al., 2019; Foa, 2010; Meuret et al., 2012; Rothbaum & Schwartz, 2002; Telch et al., 2014). In line with the CC framework, ET assumes that the disorders are caused by learned associations between an undesired behavioral responses (UR) and one or many triggers (CS), which in turn elicit the same maladaptive behavior (CR). The goal of ET is to reduce or extinguish the maladaptive behavior. Therefore, first the triggers of maladaptive behavior need to be identified, if they are unknown to the patient. Next, the patient is repeatedly exposed to the triggers in a safe environment, while avoiding pairing the triggers with the US, thus extinguishing the CR or maladaptive behavior. While ET proved to be an effective treatment (see e.g., Abramowitz et al., 2019; Craske et al., 2008; Foa, 2010; Rothbaum & Schwartz, 2002; Telch et al., 2014), a resurfacing of the maladaptive behavior may occur after ET treatment (Craske et al., 2008; Levy et al., 2021). Likewise, within CC experiments the return of the CR is an observed phenomenon counteracting the goal of ET (Gershman et al., 2013; Rescorla & Heth, 1975).

Thus, to enhance treatment efficacy many approaches or protocols on how to conduct ET evolved. Abramowitz et al. (2019) provide a broad introduction into ET, how

it is applied in practice, and which techniques are used for the variety of anxiety disorders. On a high level, first the individual's disorder and its history are assessed by the therapist. Then, an individually tailored treatment plan is developed on how to confront the disorder triggering stimuli. Exposure can be conducted in four ways: in vivo when presented live by the therapist, as stimuli imagined by the patient, in virtuo when stimuli are presented in a virtual reality (VR) environment, or interoceptively when the therapist induces the physiological responses of the disorder without presenting the CS. In cognitive behavioral therapy (CBT), ET is often accompanied with additional measures (e.g., establishing coping strategies (Meuret et al., 2012)), helping the patients to manage their disorders and to minimize their negative effects on everyday life. See Craske et al. (2014) for case studies of ET performed on patients with a variety of clinical conditions.

Important differences across treatment protocols include the intensity of the CR that is followed by the therapist's trigger exposure, the trigger exposure time, the varying similarity between used triggers, and the contingency with which the CS predicts the co-occurrence with the US and the resulting CR. For instance, flooding involves selecting the CS that triggers the patient's highest intensity response and exposing the patient to it until the behavioral response ceases. Another protocol constructs a hierarchy of the lowest- to highest-intensity triggering stimuli and exposes them to the patient in a sequence that gradually increases the patient's response intensity. In inhibitory learning a new association is formed between the same CR trigger and a new stimulus (or the absences of the previous US). This new association predicts an outcome different from the original one and competes with the previous association which outcome should be more likely predicted. Therefore, inhibitory learning aims to weaken the previous association and to reduce the contingency by which the occurrence of the CS indicates the occurrence of the US and the resulting outcome (Craske et al., 2008, 2014).

2.3 Reinforcement Learning Enabled Therapist

Importantly, the decision of which stimulus to show, the target CR intensity, the exposure time, the variation in stimulus similarity, and the contingency with which a CR follows could be controlled not only by a human therapist, but also by an RL agent making these decisions. In essence, an RL agent functioning as an RL therapist seeks to find an optimal policy for a finite Markov decision process (MDP), a mathematical formalization of a sequential decision-making problem, wherein the next state and reward depend only on the current state and action, but not on the prior history (Sutton & Barto, 2018), and where transition and reward functions are unknown to the agent. In this context, the agent is the decision-maker that interacts with an environment, a representation of everything that is external to the agent, in discrete time steps $t = 0, 1, 2, \dots$. At each time step t the agent observes a state out of a finite set of states, $s_t \in \mathcal{S}$, and chooses an action out of a finite set of actions, $a_t \in \mathcal{A}$. After each action a_t , the environment transitions to state s_{t+1} and generates a reward r_{t+1} according to the probability distribution $P(s_{t+1}, r_{t+1} | s_t, a_t)$. The agent chooses its action according to its internal policy $\pi(a_t | s_t)$, which determines its probability of choosing action a_t given state s_t . The agent's goal is to find an optimal policy $\pi^* = \arg \max_{\pi} \mathbb{E}_{\pi}[G_t | s_t]$, where the expected cumulative reward is $G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$ with future rewards being discounted with $\gamma \in [0, 1]$. For a comprehensive introduction to RL, see Sutton and Barto (2018).

RL seems promising for advancing automated or assisted ET tailored to individual patients. In the above characterization the RL agent can be seen as a model for a therapist, who, in each subsequent therapy step, needs to choose an optimal therapeutic action for their patient. More precisely, the decisions by the therapist, when to present a trigger that elicits an anxiety CR and its intensity, or the contingency with which the trigger predicts co-occurrence with the US, and other trigger parameters are equivalent to action a_t and could potentially be performed by an RL agent (RL therapist). The patient, their anxiety disorder history, and all other factors that impact the anxiety disorder can be

seen as the environment in state s_t . Notably, in reality some of these factors might be unknown or partially unobservable, making it a Partially Observable Markov Decision Process (POMDP). Hence, the MDP used in this thesis is a simplification to enable initial experiments. After each therapeutic action a_t the patient transitions from one state s_t to another uncertain state (s_{t+1}) and depending on the therapeutic outcome the RL therapist receives a reward (r_{t+1}) (e.g., reduction in the patient’s perceived anxiety). Given that the RL therapist must learn the optimal policy despite uncertainty about the best protocol and only partial observability of the true state, the RL agent must infer the state from observations. Thus, the above-mentioned MDP and the interaction with the RL agent can be viewed as an abstraction of the ET process, with a therapist (agent) interacting with a patient.

Learning in RL requires exploration (Sutton & Barto, 2018). To maximize its expected cumulative reward the agent needs to exploit the optimal action $a_t^* = \pi^*(s_t)$ at each state s_t according to its optimal policy π^* . However, usually the agent is not aware whether it has found the optimal policy. Thus, the agent needs to explore the consequences of unknown state-action pairs to potentially improve its policy. Consequently, learning in RL is typically characterized by iteratively updating π to converge towards the optimal policy π^* . The policy update is guided by the trade-off between exploiting known state-action pairs and exploring new ones which could potentially improve the agent’s policy. The agent wants to maximize the value of its policy starting at state s_0 . Hence, the agent maximizes the expected cumulative discounted reward according to the state-value function $v_\pi(s_0) = \mathbb{E}_\pi[G_t|s_0]$. Likewise, $q_\pi(s_t, a_t) = \mathbb{E}_\pi[G_t|s_t, a_t]$ denotes the action-value function, which quantifies the expected cumulative reward of taking action a_t in state s_t and following policy π afterwards. Both functions are usually unknown to the agent and are approximated throughout interaction.

In on-policy RL, the agent improves the current policy π that it uses to interact with the environment and generate trajectories (Sutton & Barto, 2018). In other words,

the same policy is used both for action selection and for policy optimization, using data generated by that same policy. At the same time, it restricts learning to trajectories generated by the current policy, which can be a disadvantage in healthcare applications where exploratory actions may be unsafe for patients. Moreover, when data are scarce and costly to obtain, it may be difficult to learn good actions for states that are rarely reached or difficult to access under the current policy π . Finally, on-policy learning cannot directly reuse trajectories collected under past or alternative policies to improve the current policy.

In contrast, off-policy RL can learn from data generated by other policies or agents. In off-policy RL two separate policies are maintained, the target policy π and the behavior policy b . While the target policy is the one that is optimized, the policy b determines the agent’s behavior during learning. Policy b covers the same action space $\mathcal{A}$, but typically exhibits a more exploratory behavior than the target policy, and as a result different state and state-action value function estimates, $v^b(s)$ and $q^b(s, a)$. Some off-policy methods also learn from stored past state-action-reward transitions generated by past or alternative policies. This flexibility is particularly relevant in ET, where state-action-reward transitions corresponding to patient state, therapeutic intervention, and disorder improvement collected under one treatment protocol may still inform learning under another. In the long term simulating patient behavior by patient models might not suffice for learning professional and clinically safe to apply ET. At some point it might be beneficial to let the RL therapist interact with real patients or at least to use real human patient data.

In these cases, off-policy RL offers important advantages. Firstly, in scenarios involving human patients, data collection may be ethically constrained and costly to acquire. Off-policy learning allows reusing scarce data by learning from stored state-action-reward transitions. Secondly, it allows improving the RL therapist’s treatment policy π from past data that was obtained through other treatment policies, or through other (RL) therapists’ interactions with patients. This enables policy or treatment protocol improvement from past data that was generated by a variety of treatment protocols that

are widespread in professional ET (e.g., flooding, or gradual trigger exposure), without needing to be applied by the RL therapist. As such, the risk of exhausting patients with potentially non-effective actions or incorrect treatment decisions is mitigated. Thirdly, exploring actions that lead to extremely high anxiety states or other therapeutic actions that might potentially harm the vulnerable patient would be controversial and likely not approved by an ethical commission. However, to improve its policy the RL therapist would tend to explore as many state-action pairs as possible. To avoid harming the patients by exploring potentially dangerous state-action pairs, off-policy RL enables learning from stored state-action-reward data. The data could stem from simulations with patient models and would effectively reduce the risk of taking potentially harmful actions, thus making RL-assisted ET safer to conduct.

DQN-algorithm

DQN (Mnih et al., 2013) is one of many RL (off-policy) algorithms that tries to find an optimal policy for a given MDP. It maximizes its expected cumulative discounted reward by learning an approximation of the optimal Q-value function

$$Q^*(s, a) = \max_{\pi} \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \mid s_t = s, a_t = a, \pi \right] \quad (20)$$

using a deep neural network in a stationary MDP environment. By maintaining separate target and behavior Q-networks, DQN stabilizes learning. The policy used by the agent to interact with the patient model and to generate the data is derived from the behavior Q-network, that maps the state vector to actions. The target Q-network is a copy of the behavior Q-network, updated periodically or via soft updates with factor $\tau \in [0, 1]$ at each training step, and is used to compute stable targets for training the behavior network that ultimately yields the optimized policy. To allow exploration during training the algorithm chooses random actions to interact with the environment with probability $\epsilon \in [0, 1]$.

Learning is stabilized by removing correlations between subsequent interaction steps. In DQN, that is achieved by uniform random sampling of mini-batches from a replay memory

buffer (Mnih et al., 2013), which stores interaction tuples $e_t = (s_t, a_t, r_t, s_{t+1}, d)$, where d indicates whether the full episode was completed, s_t being the starting state, and r_t the resulting reward of the respective interaction. The experiences are stored in a dataset called $D = \{e_1, \dots, e_t\}$. By storing older interactions from different policies, the buffer can facilitate learning in sparse reward settings (Fedus et al., 2020). Similar to many RL algorithms, DQN approximates the optimal Q-value function through iterative updates every C steps by minimizing the loss function

$$L_i(\theta_i) = \mathbb{E}_{(s,a,r,s') \sim U(D)} \left[\left(r + \gamma \max_{a'} Q(s', a'; \theta'_i) - Q(s, a; \theta_i) \right)^2 \right], \quad (21)$$

where θ_i are the weights of the behavior Q-network at iteration i , θ'_i are the weights of the target Q-network, $Q(s, a; \theta) \approx Q^*(s, a)$, and the subscript i denotes the i -th training iteration. These weights are the parameters of the deep neural network that DQN uses to approximate the Q-function. The behavior Q-network is directly optimized by using gradient descent on the loss function, defined as the mean squared difference between the target, which is the sum of the obtained reward and the discounted target Q-value, and the Q-value prediction of the behavior Q-network. For additional stability, the target Q-network is updated only after a predefined target update interval (every C steps). The algorithm is described in Algorithm 1.

Sim2Sim

Particularly in the sensitive area of healthcare and ET, it is crucial that an RL agent performs ET or EL safely and robustly across diverse patients and clinical contexts. That is, the optimized treatment policy should generalize beyond the specific patient model and conditions used during learning EL from simulations and remain effective during previously unseen dynamics or parameter variations. One way to facilitate transfer to real-life application is to model the simulated environment as realistic and representative of real patients as possible. Another way to achieve robust transfer is called Simulation-to-Simulation (Sim2Sim) or model mismatch robustness (Gordon et al., 2019).

Algorithm 1 DQN with Experience Replay (Mnih et al., 2013)

Initialize replay memory buffer D with capacity N Initialize action-value function $Q(s, a; \theta)$ with random weights**for** $episode = 1, \dots, M$ **do**Initialize state s_1 **for** $t = 1, \dots, T$ **do**With probability ϵ , select a random action a_t Otherwise, select $a_t = \max_a Q^*(s_t, a; \theta)$ Execute a_t and observe reward r_t and next state s_{t+1} Store transition $(s_t, a_t, r_t, s_{t+1}, d)$ in D Sample a random minibatch (s_j, a_j, r_j, s_{j+1}) from D

Set

$$y_j = \begin{cases} r_j, & \text{for terminal } s_{j+1}, \\ r_j + \gamma \max_{a'} Q(s_{j+1}, a'; \theta), & \text{for non-terminal } s_{j+1} \end{cases}$$

Perform a gradient descent step on $(y_j - Q(s_j, a_j; \theta))^2$ according to differentiated

Equation 21

end for**end for**

In Sim2Sim, the RL agent is trained in one source simulator and evaluated in a distinct target environment (Cobbe et al., 2019), with the goal of assessing whether the learned policy is robust to changes in environment dynamics, rather than being specialized to the particular transition and reward structure of the source training simulator (Zhang et al., 2018). The target environment may be constructed either by re-parameterizing the same underlying source simulator, thereby inducing a different MDP with altered transition dynamics, or by employing an entirely different simulation model with a modified structure, and additional parameters (Gordon et al., 2019). Another standard approach to achieve robustness is domain randomization, which samples environment parameters from

a distribution at the beginning of each training episode (Tobin et al., 2017). An episode is a complete interaction sequence between the RL therapist and the environment, that is the patient model, over the number of steps that constitute the complete EL process. Due to domain randomization the RL agent’s policy is optimized over a distribution of diverse patient models, wherein different parameter levels reflect differing patient characteristics. While domain randomization tends to increase training time, it has been shown to improve transfer to real world application (Tobin et al., 2017; Zhao et al., 2020). Zhao et al. (2020) provided an overview, that introduces a variety of methods improving RL agent’s policy transfer from the simulative training environment to reality (for more details see Section 3.3.2).

Finally, to narrow the gap between demand and availability for anxiety treatment, the RL therapist needs to be able to interact with the human patient. Practically, it is possible for the RL therapist to interact with a real patient directly or indirectly. In a virtual reality exposure therapy (VRET) session the RL therapist could be integrated into a software that decides which stimuli would be appropriate to show when, and automatically display them through VR devices. A variety of VR environments can depict various anxiety triggering situations like a social interaction (Emmelkamp et al., 2020), different heights (Rimer et al., 2021), spiders (Donker et al., 2019), or a combat situation (Beidel et al., 2019). In these VRET sessions the psychophysiological response of the patient could be measured (e.g., skin conductance, pupil size, or heart rate), from which the RL therapist could infer the level of the patient’s CR and adapt its therapy accordingly (Bach & Melinscak, 2020; Rahman et al., 2022). Moreover, indirect interaction as a guiding assistant to a human therapist is possible, too. That is, the RL therapist can provide the human therapist decisions on which triggers to display when, or to assess which CR level are required at each time step, or by adapting the stimulus to be shown contingent on the anticipated CR level. A number of systematic reviews showed VRET to have no significant difference in efficacy compared to traditional therapy. VRET was found to be an efficacious

approach to treat anxiety related disorders (Carl et al., 2019; Powers & Emmelkamp, 2008), and post-traumatic stress disorder (PTSD) (Deng et al., 2019; Gonçalves et al., 2012). Thus, VRET is not only effective but also ideally suited to integrate RL with ET and to assist with ET.

3 Related Work

There are three interconnected threads of literature which are relevant to automate ET with the help of an RL therapist. The first thread researches and models associative learning. The second thread researches ways to improve ET and contains translational studies to incorporate advances from associative learning theories into clinical practice. The third thread advances the discipline of reinforcement learning and its applications in healthcare. Lastly, works which combine all three threads to automate ET are discussed.

3.1 Modeling Associative Learning

Currently, two main ideas are guiding research in associative learning (see Dunsmoor et al., 2015, for review). The RW model and its proponents follow the idea that extinction is caused by weakening the learned associations until they are deleted, or erased from memory (Rescorla & Wagner, 1972). In contrast, the Pearce-Hall model (Pearce & Hall, 1980), conceptualizes extinction as the formation of a new inhibitory memory, which competes with the previously acquired association, determining whether to cause a CR. Although acknowledging the requirement of further integrative research, Bouton et al. (2021) reviewed behavioral and neuro-biological studies and concluded that there would be stronger support for the formation of a new inhibitory memory. While RW performed well in approximating learned associations and estimating their strength, it came short in explaining empirical findings like the return of the CR and other empirically observable phenomena (Miller et al., 1995). If an association between a cue and its US was truly unlearned, a return of CR measured by a sudden recovery after days (Quirk, 2002), after a short reinstatement (presenting the US alone, Haaker et al., 2014), or after renewal (reintroducing the CS-US pairing in the original acquisition context or a novel one, distinct from the extinction context, Vervliet et al., 2013) would be inconsistent with the predictions of RW (Gershman et al., 2015).

Reconciling, the group of latent cause models (LCM) (e.g., Courville et al., 2003, 2006; Gershman et al., 2014, 2015, 2017) integrated both ideas with the observation that

the context during extinction might have a crucial impact on which idea holds true (Bouton, 2004). In LCMs the learning becomes probabilistic and is comprised of two processes, the associative learning, and structure learning (Gershman et al., 2017). Structure learning postulates that learners attempt to minimize the uncertainty and complexity of their surroundings by inferring latent causes, that are presumably causal for the surroundings. In basic LCMs the learner observes sensory input, such as a CS or a US, and other contextual features. Each time step the learner infers a latent cause through a posterior distribution over potential latent causes using Bayes rule, based on the sensory input and prior learning. With each time point a new latent cause can be inferred and generated, when the previous one seems unlikely to hold. Nevertheless, a prior probability distribution favors few latent causes (Gershman et al., 2015). The observed input is evidence to the learner’s inference of the active latent cause. With two distinct latent causes, the Bayesian learner can maintain two context dependent CRs for the same pair of CS and US (Gershman et al., 2015). The CR occurs when the learner expects the occurrence of the US under the inferred latent cause. A change in the context even when the CS remains present, shifts the calculated posterior probability distributions over the latent causes, and with it the US prediction.

Hence, whether a learned CR is removed from memory or rather out competed by a new inhibitory memory depends on the context and the inferred latent causes. Theoretically, an LCM allows a return of CRs after the extinction phase to be consistent with its predictions (Gershman et al., 2015). If during the extinction phase the learner infers a new latent cause due to context shift (e.g., sudden absence of the US) the association remains in memory but is not exhibited anymore due to a change in the inferred active latent cause. In this case the account of the new inhibitory memory holds true. However, if the learner has no reason to infer a new latent cause, the true unlearning of the CR remains feasible. The acquisition phase and the extinction phase need to be as similar as possible to avoid inferring new latent causes. While its hard to control similarity

of contexts in real world scenarios, in experimental settings it is feasible. Introducing new stimuli or omitting previously shown stimuli diminishes similarity and can cause the inference of new active latent causes. Thus, introducing gradual extinction increased the similarity to the acquisition phase and resulted in an effective erasure of the CR return in past research (Gershman et al., 2013; Morís et al., 2017; Shibani et al., 2015; van den Akker et al., 2015). Gradual extinction can be achieved by starting with trials where CS and US are shown concurrently, and by gradually decreasing their frequency towards the end of extinction phase.

On a computational level the RW model needs to be adapted to allow for probabilistic Bayesian associative learning (see Kruschke, 2008, for an intro to Bayesian learning). The Kalman filter, a Bayesian generalization of RW, was originally developed as an iterative approach to estimate unknown parameters in dynamic systems in the field of engineering (Kalman, 1960). Nevertheless, it proved to be beneficial when applied to associative learning (Dayan & Kakade, 2000; Kakade & Dayan, 2002) and resulted in the Kalman-Rescorla-Wagner model (KRW, see Section 2.1 for mathematical details) of associative learning. The three most important changes to RW are firstly, that the prediction of the associative strength turns into a prediction of a probabilistic distribution of the associative strength, instead of predicting a single point. With increasing uncertainty over time, the associative strength can shift due to random diffusion of the cue weights (Gershman, 2015; Kruschke, 2008). Secondly, the cue weights might be dependent on each other. A covariance matrix captures potential correlations, and reflects belief uncertainty over the nature and associations of the cues (Kruschke, 2008). Thirdly, with increasing availability of observations, the learners beliefs about the association are strengthened and the learner’s degree of uncertainty diminishes (Gershman, 2015).

Significantly, KRW succeeds in modeling backward blocking and latent inhibition, two empirical phenomena that RW failed to predict (Miller et al., 1995). Backward blocking refers to the observation, that when two cues A and B are paired in compound

with a US, and subsequently learning continues only by pairing cue A with US, observing cue B leads to lower CRs than when tested immediately after the compound learning phase (Kruschke, 2008). In other words, cue B is retrospectively re-evaluated, although not being present at learning trials. While in RW all cue weights are independent and the predictive power of a cue does not change if not presented in learning trials, in KRW it does. Beside random diffusion of weights over time, in backward blocking KRW learns that both stimuli are to some degree redundant in producing the CR, and thus that the weights are summing up to one with a negative covariance matrix. Therefore in line with backward blocking, an increase in the weight of cue A due to continued presentation necessarily causes a decrease in the weight of cue B (Gershman, 2015). Latent inhibition occurs when observing the NS alone without the US prior to the CR acquisition phase. In subsequent trials, it slows down learning the association between the NS and US. While in RW no learning occurs when an NS is presented alone, in KRW the pre-exposure of NS decreases the observation noise about the nature and associations of NS and causes slower association learning (Gershman, 2015).

Furthermore, the KRW model can be extended with the structure learning component of LCMs. In LCMs making sense of the surrounding is reduced to the Bayesian inference over latent cause assignment. One parsimonious and computationally fast way to account for this dynamic when observing behavioral data is to add a Dirichlet-Process to the KRW model (DPKRW, Gershman et al., 2014, see Section 2.1 for mathematical details). The DPKRW, and LCM in general, were used as a theoretical model for the updating and formation of new episodic memory traces in brain structures (Gershman et al., 2014, 2017). Accordingly, an inferred latent causes corresponds to a distinct memory trace that evolves over time. Its theory proposes, that the memory system in the brain tracks the sensory input, and estimates probabilities distributions as a Bayesian learner. Each inferred latent cause is the result of different posterior probabilities of weight vectors. The Dirichlet-Process enables to generate an infinite amount of latent causes, while the

prior probability distribution and stickiness parameter causes a bias towards simplicity by inferring fewer and already established latent causes (Gershman et al., 2014). New latent causes tend to be inferred from abrupt salient changes of context, stimulus novelty, the magnitude of the prediction error, or temporal and spatial distances and discontinuities (Gershman et al., 2014).

3.2 Translating Experimental Findings into Exposure Therapy

The second thread of literature concerns with the question on how ET treatments can incorporate the theoretical considerations and empirical findings into clinical practice to benefit the patients. Firstly, LCM’s implication that the context during extinction should be as similar as possible to acquisition proved to be beneficial for improving ET. Studies which adopted a gradual extinction protocol, to increase context similarity, obtained lower relapse rates of CRs than in standard extinction procedures (Morís et al., 2017; Shiban et al., 2015; van den Akker et al., 2015). In gradual extinction protocols US stimuli are occasionally paired with the CS during the extinction phase. The frequency of the pairing gradually decreases over trials. Whereas, in fixed gradual extinction protocols the trials on which pairings occur are predetermined (Gershman et al., 2013; Shiban et al., 2015), in stochastic protocols the probability of a pairing decreases over the extinction phase (Morís et al., 2017; van den Akker et al., 2015). The effect of gradual extinction protocols is in line with the occasional reinforced extinction strategy, wherein occasional reinforcement might slow down EL as a trade-off, but at the same time attenuate subsequent reacquisition or relapse of the CR (Craske et al., 2014). Since LCMs are relatively newer than the inhibitory memory account of extinction, translational studies more commonly focus on the latter (e.g., Carpenter et al., 2019; Pittig et al., 2016; Zlomuzica et al., 2020).

Secondly, the introduction and maximization of expectation violation can enhance extinction in ET (Craske et al., 2014; Pittig et al., 2016). Essentially, the idea is to adapt ET to the patient’s individual expectations of US occurrence and to dis-confirm them in the strongest possible way. One way to maximize the expectation violation when the

patient expects the US to co-occur with the trigger or CS due to its anxiety disorder condition would be to present the trigger in absence of the US. Once the patient learned that the trigger will be shown alone during ET, it might be beneficial in terms of expectation violation to occasionally pair it with the US. Abrupt variation of exposure time of one of the stimuli or introducing against expectations new NS might further maximize expectation violation. Commonly, expectation violation is considered to be a key driver of associative and extinction learning. Furthermore, it might play an important role in the formation of a distinct inhibitory association between the anxiety trigger and the absence of the US, since due to initial expectation violation a new memory trace might be inferred.

Thirdly, reducing context and stimulus specificity to foster broader generalization of extinction during ET can be achieved by varying the used triggers, contexts, and inter-exposure intervals (Craske et al., 2014). Deepened extinction involves multiple triggers that invoke an equivalent fear CR, are first used for extinction training separately, and afterwards for combined ET (Craske et al., 2014). For instance, in ET for spider phobia different interchanging spiders are used for ET, before they are exposed in combination. Likewise, it was found that the variation of time intervals between distinct exposure offsets had a positive impact on the long lasting extinction (Craske et al., 2014). Varying the exhibited CR levels, by means of varying the exposure time or triggers, had similarly positive effects on long-term extinction (Craske et al., 2014). Furthermore, it was observed that the contexts of fear CR renewals deviated from the context during ET. Thus, in order for the patient not to not learn context dependent extinction, the context during ET should be varied (Craske et al., 2014).

Finally, Carpenter et al. (2019) suggested experimental adaptations for extinction research to better generalize into clinical settings of phobias and capture their complex nuances. Accordingly, fear relevant CS (e.g., picture of angry face) should be used instead of NS (e.g., geometrical shape), or the CS should vary within its category (e.g., snake) across trials to better reflect clinical generalizations of fears. Another important aspect of

translating extinction learning research into clinical practice is to consider differences in learning among various patients. A systematic review of computational psychiatry studies on mood and anxiety patients revealed reduced reward and increased punishment learning rates for patients compared to control groups (Pike & Robinson, 2022). This finding further backed up using fear relevant stimuli in research and suggested that patients acquired clinically relevant CRs faster than extinguishing them, independent of whether the inhibitory or unlearning account holds true. Thus, realistic patient parameter estimates might enhance extinction research and facilitate its translatability into clinical ET. Moreover, experiments should reflect cognitive processes involved in CBT which support fear extinction. For instance, cognitive reappraisal refers to the conscious attempt by the patient to adopt an alternative perspective on the fear triggering stimuli, or mental imagery is the attempt to imagine a different outcome than the UR, and thus to create a new relation between the CS and its outcome.

3.3 Reinforcement Learning in Healthcare

The third thread researches and advances the application of RL in healthcare in general, and off-policy, Sim2Sim, or RL-assisted ET in particular. Past RL applications in healthcare can be categorized into precision medicine, dynamic treatment regimes, personalized rehabilitation, medical imaging, diagnostic systems, control systems, dialog systems, and health management systems (Coronato et al., 2020). Across categories past works used predominantly Q-learning, followed by Deep-Q-Network (DQN) (Mnih et al., 2013, 2015). Both RL algorithms are off-policy methods that seek to find an optimal policy in MDPs. Both methods together account for more than half of the RL applications in healthcare research (Coronato et al., 2020). One example of off-policy RL in healthcare is the AI Clinician that produced post-hoc sepsis treatment decisions on retrospective data (Komorowski et al., 2018). The found treatment policy was evaluated against past clinical decisions, which served as the behavior policy. It was demonstrated that the AI Clinician's found policy reliably outperformed the clinician's behavior policy. Patients that received

medication doses that resembled the AI Clinician’s policy the closest showed the lowest mortality rate. The AI Clinician demonstrated the feasibility of optimizing treatment decisions, and find a policy that may assist clinicians to improve treatment outcomes.

3.3.1 Off-Policy Evaluation in Healthcare

Nevertheless, the publication of the AI Clinician sparked a debate on off-policy in healthcare and led to a number of methodological suggestions for improvement (Gao et al., 2023; Gottesman et al., 2018). RL in healthcare needs to conform to high safety standards. Therefore, evaluating RL in healthcare studies should consider, whether the effective sample size was sufficient, whether all long-term effects of treatments were adequately considered, and whether all potential confounding variables used by human clinicians for their corresponding decisions were available to the RL agent (Gottesman et al., 2019). The sample size denotes the amount of patients or interactions that were used for evaluating the learned policy of the RL agent. A small sample size may not be representative or biased and distort findings about the found policies or the RL agent. Yet, a smaller effective sample size than stated was often the consequence of deterministic policies with importance sampling that were used for off-policy evaluation (Gottesman et al., 2018). A deterministic policy exploits the optimal policy it has found and ceases stochastic exploration of state-action pairs entirely. Importance sampling is an off-policy technique to sample state-action pairs under behavior policy b and from those to estimate the expected cumulative reward under the target policy π . However, if policy b produces actions that are not produced by the deterministic target policy, these actions are discarded due to their zero weight assignments. Therefore, evaluating the target policy only against the behavior policy, which often is comprised of the empirical clinical data, might be misleading and uses an effectively lower sample size than the one stated (Gottesman et al., 2018). In sepsis treatment potentially misleading results and harmful target policies were found by the off-policy RL agents, that combined deterministic policies with importance sampling for evaluation (Gottesman et al., 2018).

Moreover, the practice of comparing the treatment policy against an expert clinician behavior policy (with past data), might altogether fail to provide a meaningful benchmark, when not controlled for confounding variables or expert domain knowledge (Gottesman et al., 2018). More precisely, associating the deviation of recommended medication dosage to the one administered with an mortality rate outcome in sepsis treatment, might be more informative on the seriousness of the patient’s state instead of the utility of the recommended policy. In fact, a random action policy and a no action policy produced a similar association with mortality curve as did the recommended policy (Gottesman et al., 2018). Both the random and no action policies are known to be harmful to patients suffering from sepsis. Likely the curve reflected the state of the patient, and that patients in more serious states required higher medication dosage and were more likely to suffer from a fatal disease outcome. Likewise, patients that were in a less serious state but had received high medication dosages were likely to suffer fatal outcomes from overdosing. Both shortcomings, the importance sampling and the expert clinician benchmark, were used for evaluating the AI Clinician, and thus the evaluation remained inconclusive.

Training and evaluation of RL in healthcare is further complicated by sparse, aggregated rewards, and by only partially observable patient states (Gao et al., 2023). Often the desired outcome of the policy is quantified as a final reward (e.g., fear extinction after ET, or survival after sepsis treatment), with sparse or no observed rewards along the treatment process. However, zero rewards complicate learning of effective state-action pairs, and impede their evaluation, since it is not clear what state-action pairs can be credited to what amount for obtaining the final non-zero reward. In addition, the patient state cannot be fully observed and must be estimated by an observation function. An only partially observable MDP (POMDP) is shown in $(\mathcal{S}, \mathcal{A}, P, \mathcal{O}, \Omega, R, \gamma)$, where the newly added $\imath \in \mathcal{O}$ is the observation space of the agent and $\Omega = p(\imath|s)$ is the observation function by which the agent estimates the state parameters (compare with Section 2.3). Addressing sparse rewards and the POMDP properties in healthcare applications, a human-centric

off-policy evaluation method (HOPE) was introduced (Gao et al., 2023). In a sepsis treatment study HOPE addressed both challenges by inferring non-zero reward values based on the final cumulative reward and state nearest neighbor imputation. Importantly, HOPE managed to outperform other standard off-policy benchmark evaluation methods for RL policies (e.g., importance sampling). Nevertheless, HOPE still applies the logic of comparing the learned treatment policy against an expert clinician behavior policy. Overall off-policy evaluation was found to poorly predict performance when applied to another environment (Hargrave et al., 2024).

3.3.2 *Sim2Sim*

For RL-assisted ET to succeed it must be safe to apply. To ensure safety RL has to perform reliably, stable, and its learned policies must generalize well across unseen contexts or patients. Sim2sim, or mismatch robustness are a group of techniques that seek to ensure a stable and robust application of learned policies in real (Sim2Real) or other simulated (Sim2Sim) (Gordon et al., 2019) environments that are different to the one on which the RL agent was trained. Most intuitively, one approach is to model the learning environment as realistic and as similar to the target environment as possible (Zhao et al., 2020). This strategy is called zero-shot and it is supposed to allow direct application of the learned policy in the target environment. However, it is often difficult to model the simulative environment representative of the target environment in healthcare. In particular, it is challenging to accurately model POMDPs like ET, where subsets of the (target) environment are not observable, and where research gaps remain to be studied. Zero-shot strategies are often accompanied by domain randomization, the idea that randomly sampling all learning environment parameters might cover realistic parameter configuration distributions, that are present in the target environment (Tobin et al., 2017; Zhao et al., 2020). In fact, domain randomization was widely observed to lead to more robust RL policies in return for slower learning (Tobin et al., 2017; Zhao et al., 2020). Another method is the already above discussed and discrepancy-based off-policy evaluation that

contrasts learned policy’s decisions with expert decisions from another behavior policy. In another approach for coordinated multiple RL agents the entire environments used for learning in robotics settings were perpetuated to allow learning from deviating environments (Tobin et al., 2017).

Despite the grave need for safe and robust RL methods for clinical applications, healthcare specific RL robustness research is rare, and most techniques are researched and sourced from unrelated fields (e.g., robotics, or computer vision). Nonetheless, three notable exceptions were recently published. Firstly, cross off-policy evaluation introduced varying weights of single rewards during policy evaluation compared to the weights used during training (Roggeveen et al., 2024). The rationale was that clinically beneficial and robust policy would perform well independent of the used reward structure during training. In addition the policy was restricted to not allow malicious actions during any state. Secondly, with EpiCare a simulative environment benchmark was introduced to perform Sim2Sim and robustness evaluations of RL agents applied in dynamic treatment regimes (Hargrave et al., 2024). Dynamic treatment regime is an abstraction of ET, and characterize any treatments, where parameters like symptoms, intensity, or fear levels change, and thus require patient specific treatment adaptation over time. Evaluation with EpiCare revealed poor performance with high variability of RL agents in sparse data scenarios with limited observability, despite a seemingly good performance with off-policy evaluation. DTR-Bench introduced an simulation platform to model realistic patient responses and dynamic treatment regime dynamics to benchmark RL agents in Sim2Sim settings before evaluating potential performance transfer to real patients (Luo et al., 2024).

3.3.3 Automating EL with RL

The most closely related works combined the above-mentioned threads of literature to automate EL or ET (e.g., Mahmoudi-Nejad et al., 2021, 2024; Rahman et al., 2022; Tomasiak, 2024). A core challenge on the way to automated ET is to provide the RL therapist with real-time (bio-)feedback from patients or their observed CR. Rahman et al.

(2022) proposed a framework to integrate real-time data into VRET interventions. Using electroencephalography, they measured cortical activity and combined it with heart rate variability to predict patients' arousal or stress levels. They successfully demonstrated the feasibility of providing real-time feedback that an RL therapist can use to form treatment decisions in VRET. The next step toward automated ET would be to form automated treatment decisions, such as determining which stimuli should be presented, based on patient feedback.

To this end, Tomasiak (2024) developed an RL therapist that interacted with an artificial patient model. Based on the patient's observed CR, the RL therapist decided which stimuli to present with the goal of reducing the CR. An important finding was that EL training worked best when delivered over episodes with EL 24 interactions, yielding greater reductions in CR than longer treatment periods. Complementarily, Mahmoudi-Nejad et al. (2021) introduced a framework in which an RL agent dynamically generated spider content for VRET sessions to elicit targeted levels of arousal or stress. By varying six stimulus dimensions (locomotion, movement, proximity, hairiness, color, and size), they were able to produce over 100 distinct spider stimuli and model patient arousal as a function of stimulus configuration.

Together, these studies seemed promising for applications in VRET. An RL therapist could use real-time biofeedback from patients to estimate their CR levels, and form treatment decisions accordingly. Subsequently, it could generate stimuli designed to induce a CR of the desired intensity, tailored to differing individuals, and depending on the chosen therapy strategy or the severity of the disorder. Importantly, the last two studies relied on artificial patient models. However, in a follow-up study, Mahmoudi-Nejad et al. (2024) demonstrated that dynamically adapted spiders elicited higher anxiety levels in human study participants compared to a calming condition. Moreover, they showed that spider configurations could be tailored to elicit predetermined anxiety levels, and that the spider stimulus to achieve the desired level varied across individuals, thus supporting the

personalization of spiders to individuals. Notably, their participants were not clinically arachnophobic, and the study did not aim to assess therapeutic outcomes. Instead, it provided a proof of concept that RL based adaptation could generate stimuli capable of inducing specific anxiety responses. Thus, fully automated ET has not been studied yet and its feasibility and therapeutic efficacy remain open questions.

4 Methodology

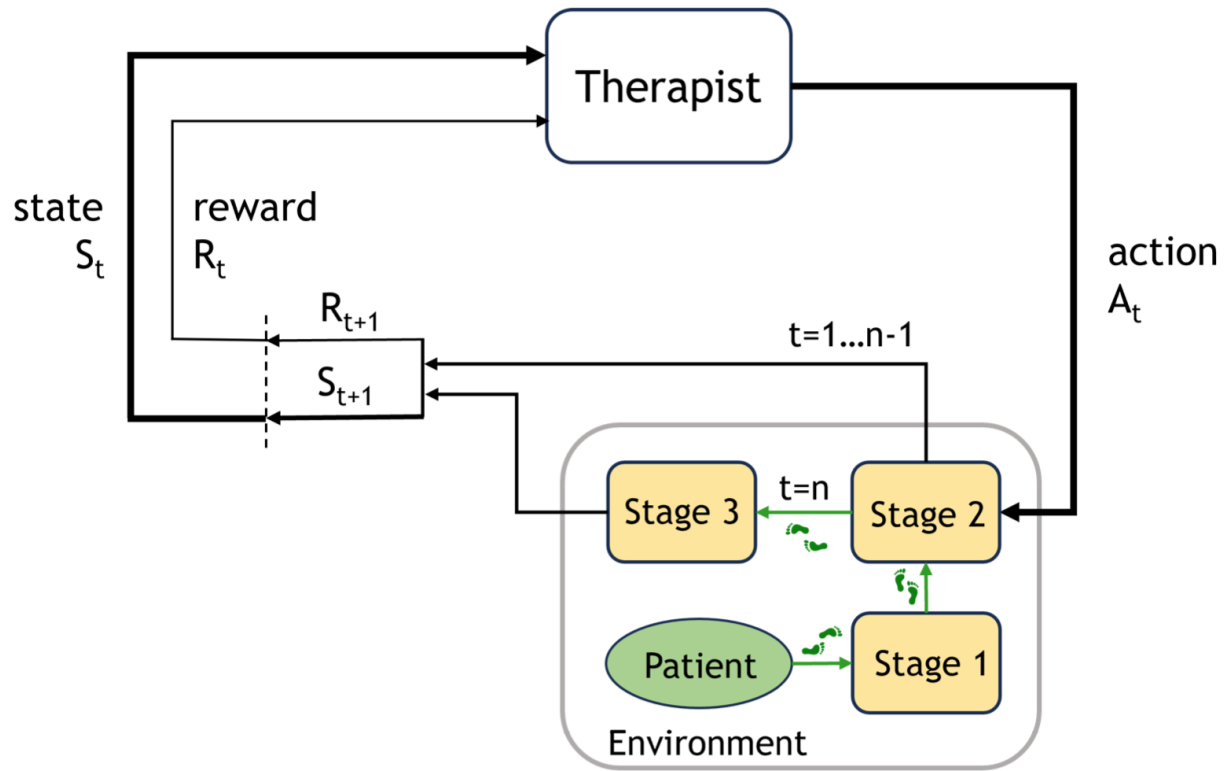
The goal of this thesis is to advance the research on the path to automated, RL-assisted ET. First, a broad overview of the experimental setup is provided to introduce the key components. Subsequently, each component is described in detail in dedicated subsections.

4.1 Broad Overview

Each experiment simulated a three-stage interaction between an RL agent and a customized environment, with the objective of enabling the agent to learn how to conduct EL. The overall experimental framework can be seen in Figure 1, which shows the RL formulation of the problem. The environment was implemented using a standard *Gym* interface environment (Brockman et al., 2016), which facilitates the representation of RL problems and provides a standardized protocol for agent–environment interaction. This environment was specifically customized to capture the essential characteristics of EL in lab settings. Within the environment, study participant behavior was modeled using three distinct computational patient models: the RW (Gershman, 2015; Rescorla & Wagner, 1972), the KRW (Gershman, 2015), and the DPKRW (Gershman et al., 2017) models. The RL agent was implemented using a DQN algorithm to model the RL therapist (Mnih et al., 2013, 2015).

The EL experiments consisted of three stages. In Stage 1, the patient model was conditioned to acquire a CR. Although real patients typically acquire such responses prior to therapy, this stage is necessary in experimental settings to establish the CR and to generate variability in CR strength across simulated participants. In Stage 2, the RL therapist performed EL to reduce the CR. In stage 3, a test of CR return was conducted to evaluate the effectiveness of extinction. Strictly speaking, agent–environment interactions occurred only during Stage 2, whereas Stages 1 and 3 were controlled entirely by the environment dynamics. During Stage 2, the RL agent operated under a sparse reward regime, receiving zero reward at each interaction step. The cumulative reward was

computed only at the end of the full episode, encompassing all three EL stages, and was then assigned to the final interaction step in Stage 2. At each interaction step, the RL agent selected an action from a discrete set of six possible actions. The environment state was defined by the CR level of the patient model, which could evolve at every step across all stages. However, only in Stage 2 did a step correspond directly to an interaction between the RL therapist and the patient model.

**Figure 1**

Experimental setup of RL problem adapted to EL

Note. The overall setup of the RL problem adapted for EL is shown. The agent is the RL therapist who selects an action from a set to interact with the environment every step. The environment consists of the patient model and three stages of the employed EL paradigm, where the observable state is the patient's CR level. Only after the final patient interaction in stage 3, the RL therapist receives a non-zero reward, which depends on its efficacy in lowering the CR level. Reproduced from "Optimizing Exposure Therapy Protocols Using Reinforcement Learning", by Tomasiak (2024).

Experiments and Variables. Four experiments were conducted to address the research questions outlined above. Experiment 1 examined the effect of experience replay buffer composition, hereafter referred to as the buffer, on EL with off-policy RL agents. The buffer is used in DQN to store past interactions collected under earlier or alternative policies (see Section 5.1 for details). Experiment 2 examined the effect of the number of stored interactions on EL with off-policy RL agents. The number of interactions stored within one episode corresponds to the number of interactions conducted between an experimenter and a study participant (see Section 5.2 for details). Experiment 3 examined the effect of different patient models in the evaluation environment on the robustness of EL (see Section 5.3 for details). Experiment 4 examined the effect of parameter shifts in patient models within the evaluation environments on the robustness of EL performance (see Section 5.4 for details).

Hardware and Software. The simulations and computations were conducted on a Linux server running Ubuntu 20.04. The server was equipped with two Intel 4214R processors and two NVIDIA A100 GPUs. The simulations utilized a design and Python code adapted from Tomasiak (2024). The code is available at this [GitHub repository](#). The EL environment was implemented using a customized *Gym* environment (Brockman et al., 2016). *Gym* is a Python library that facilitates the creation of custom environments and their interaction with RL algorithms. The DQN algorithm representing the therapist was implemented using *Stable-Baselines3* (Raffin et al., 2021), which follows the DQN variant proposed by Mnih et al. (2015).

4.2 Environment: Therapy Session

The used environment was a standard *Gym* interface environment, customized to specifically model an EL lab setting in which an episode was defined as all interactions between the RL therapist and a patient model within three sequential stages. The environment consisted of the following components:

- **Patient model:** One of three patient models was possible in the environment: RW,

KRW, DPKRW (see Section 4.3).

- **Stage 1:** The patient model acquires an CR
- **Stage 2:** The RL therapists performs EL and interacts with the patient model.
Strictly speaking, stage 2 is not part of the environment, but the interaction with it.
- **Stage 3:** The effectiveness of EL is tested.
- **Action space:** Six different actions are possible, that correspond to combinations of cue A, cue B, and the US.
- **Observable state:** The patient model's CR level $\in [0, 1]$ is the state, that the RL therapist can observe.

While stage 1 and 3 were conducted automatically within the environment, stage 2 was strictly speaking not part of the environment but the interaction with it. One interaction corresponded to one environment step or one action taken by an RL therapist during training. The observable state was the CR level exhibited by the patient model, with $CR \in [0, 1]$. In each step the therapist's action was to show a set of stimuli. In the experiments there were potentially three different stimuli present, cue A, cue B, and a US. The association leading to a fear CR will be formed between cue A and a US. Cue B serves a contrasting second stimulus and can be shown by the RL therapist during EL. In conditioning experiments often more stimuli are present to introduce more complexity or model phenomena where multiple stimuli are involved (e.g., backward blocking). In the remainder of this thesis a binary vector will describe the shown stimuli, where the first number refers to cue A, the second to cue B, and the third to the US. For instance, the vector $[1 \ 0 \ 0]$ describes that the therapist has shown only cue A. Each action had to show at least one stimulus. If only a single stimulus was shown, it could not be the US. This leads to six possible sets of stimuli and, thus to six possible actions. An overview of the possible actions and other specifications of the environment can be found in Table 1. After

each step the state of the environment and the patient’s association levels between cue A, cue B, and the US changed according to the cognitive model in use.

Stage 1. The goal of stage 1, the CR acquisition, was for the patient to acquire an aversive CR for cue A, but not for cue B. Stage 1 was similar to CR acquisition in lab experiments with real human participants. See Lonsdorf et al. (2017) for an introduction into human conditioning experiments and associated challenges. Stage 1 was conducted automatically upon patient initialization, through a stochastic process but not through RL therapist interaction. It consisted out of 64 trials with only one of the cues allowed to be shown. Half of the trials (32) showed cue A. Out of these cue A was paired with a US 24 times. Cue B was always presented without being paired to any US. Thus the actions shown to the patient model were 24 times $[1\ 0\ 1]$, 8 times $[1\ 0\ 0]$, and 32 times $[0\ 1\ 0]$. All actions were randomly permuted to avoid positional effects. The last appearing cue A in stage 1 was restricted to be presented without a US to achieve a clearer CR contrast between stage 1 and 2. The rationale behind including action $[1\ 0\ 0]$, was to be able to distinguish conditioning effects. In experiments with human participants it further serves the purpose to prevent the patient from habituating to the US. Habituation may lead to decreased URs, and decreased CRs. Thus, habituation can be a confounding factor that might obfuscate the efficacy of the ET intervention. Stage 1 was triggered by default, as soon as a new patient was initialized by the environment to ensure that every patient would acquire an CR.

Stage 2. Stage 2, the CR extinction, resembled the actual ET session and its goal was to extinguish the aversive association between cue A and the US. Based on the findings of Tomasiak (2024), stage 2 consisted of 24 trials. The RL therapist decided on its own which actions out of the valid action space to perform in order to extinguish the CR. Notably, in stage 2 the RL therapist received zero reward for any action taken. The final episode reward was obtained after stage 3, which was triggered once the 24th step of stage 2 was completed. The final episode reward was then attached to the last step in stage 2.

The reward function is detailed in Section [4.4](#).

Stage 3. Stage 3, the test of CR retention, had the goal to determine whether the CR was successfully extinguished in the patient model, and the amount of reward which the RL therapist obtained for this episode depending on its success. Stage 3 was triggered after the last step if stage 2 and consisted of 10 trials. In each trial only one cue was presented at a time. In the first trial cue A was presented and in subsequent trials the alternating cue was presented. Therefore, possible actions were changing between $[1\ 0\ 0]$ and $[0\ 1\ 0]$ in each trial.

4.3 Patient: Models of Associative Learning

In the present study the patient was simulated by one of three computational models: RW (Rescorla & Wagner, [1972](#)), KRW (Gershman, [2015](#)), DPKRW (Gershman et al., [2017](#)). More precisely, each patient model simulated the evolution of the patient’s CR during interaction with an RL therapists. RW is a common and simple baseline model of human learning, which ensures comparability and interpretability of results with respect to previous research. It succeeded in explaining and predicting some of the observed dynamics of association learning. However, empirically, RW did not capture all complexities of human associative learning (Miller et al., [1995](#)). Hence, to obtain more realistic patient learning behavior estimates, KRW and DPKRW, which are both extensions of RW that succeeded in capturing more empirical phenomena, were used in addition. The models do not necessarily reflect individual differences among patients, but rather pose different learning assumptions. Inter-individual differences were accounted for by each model’s parameter configurations. As such, a robust therapy performance between the three patient models and within a patient model on with different parameter values might point towards a robust RL therapist that will be capable of tailoring EL to the needs of different individuals with different clinical or personal dispositions. The models have been described Section [2.1](#) and [3.1](#).

4.4 Reward Function and Observation Space

Reward. The reward function is an integral part of RL to obtain meaningful results on the way to automated ET. To avoid the relapse of CR, it is important that the reward function reflects long-term effects. In our simulation long-term effects refer to $CR_{\text{stage 3}}$ levels in stage 3, measured immediately after ET in stage 2, with no time gaps in between. As such, long-term effects in this thesis differ from human long-term memory effects (e.g., months after ET). The reward function in this thesis reflects the long-term effects by accounting for $CR_{\text{stage 3}}$ and contrasting them to the $CR_{\text{stage 1}}$. The reward function was taken from Tomasiak (2024), was based on Lonsdorf et al., (2019), and is shown in Equation 22, with $m = 10$ being the number of trials in stage 3 and the $CR_{\text{stage 3}}$ stands for all observed states by the agent in stage 3, that are the CRs of the patient model. $k = 10$ is the number of last trials from overall $n = 64$ trials in stage 1, of which the CR is taken into account in $CR_{\text{stage 1}}$ for reward calculation. Effectively, in the formula the mean CR level over ten trials in stage 3 was divided by the mean CR level over the last ten trials in stage 1. By multiplying the term with 100 and subtracting it from 100, a convenient measure of the CR level decrease in percentage after EL in stage 2 was obtained.

$$r = 100 - \left(100 \cdot \frac{\frac{1}{m} \sum_{i=1}^m CR_{\text{stage 3},i}}{\frac{1}{k} \sum_{j=n-k+1}^n CR_{\text{stage 1},j}} \right) \quad (22)$$

Observation Space. The observation space is the set of possible data that can be potentially observed by the RL therapist in then environment before and after a state transition or step. Ideally, it should contain all variables that therapists consider before making a decision on the treatment protocol or which stimuli to show next (e.g., age or disorder history) (Gottesman et al., (2018, 2019)). However, in this thesis for simplicity only the CR level of the patient was observed, with $CR \in [0, 1]$. After learning an association between cue A and US the patient was expected to exhibit a CR in expectation of the US to occur. In both the RW and KRW patient models the associative strength or US expectation was calculated with $v_n = w_n^T x_n$. Applying the sigmoid function to this term to

Table 1*Specification of the RL Problem and the Customized Environment*

Component	Specification
Environment	Therapy Session with three stages
Agent	DQN-Algorithm as the RL therapist
Patient models	RW, KRW, DPKRW
Action Space	[1 0 0], [1 1 0], [1 0 1], [0 1 0], [0 1 1], [1 1 1]
Observation Space	[CR] $\in$ [0,1]
Reward Function	ACQcorrectedNonDifferentialReward

model the expectation of a CR given the expectation of a US had the benefit of limiting the CR to positive values with $CR \in [0, 1]$. Otherwise, the RL therapist might learn to exploit negative weights that would lead to a sub-optimal EL outcome (Tomasiak, 2024). Equation 24 shows the CR for the RW and KRW patients:

$$CR_n = \sigma(w_n^T x_n) \quad (23)$$

$$= \frac{1}{1 + e^{-w_n^T x_n}} \quad (24)$$

For the DPKRW patient additionally the posterior probability distribution over the latent causes were considered, as shown in Equation 25:

$$CR_n = \sigma((W_n x_n)^T p_{Z_{n+1}}) \quad (25)$$

4.5 Agent: DQN as the RL Therapist

DQN (Mnih et al., 2013) was chosen as the agent resembling the RL therapist that performs EL automatically due to its frequent application in healthcare, and its off-policy ability to leverage past interaction data through an experience replay buffer. The default *Stable-Baselines3* DQN implementation with a MlpPolicy was used in this thesis. The default Q-network was a feedforward neural network with two hidden fully connected layers

of 64 units each using ReLU activation, and mapping the single-value state vector s to Q-values for the six discrete actions. For more details on to the DQN see section [2.3](#).

Experience Replay Buffer. One important experimental condition was the buffer composition (see Section [5.1](#)). In DQN an interaction tuple $e_t = (s_t, a_t, r_t, s_{t+1}, d)$ is stored in the buffer after every step. In this thesis the tuple’s state s_t refers to the patient model’s CR level, a_t refers to the DQN agent’s undertaken action, r_t is the obtained reward, s_{t+1} is the patient model’s CR level after the RL therapist presented its action, and d indicates whether that particular interaction was the final stimuli exposure by the RL therapist with a single patient. Actions are generated by the RL therapist’s treatment protocol, that is its policy derived from its Q-function, during the RL therapist’s interaction with the patient. Next to past interactions, the buffer can be filled with interactions conducted by entirely different (RL) therapists with diverging treatment protocols. The standard buffer implemented in *Stable-Baselines3* has a predefined buffer size capacity. Once the buffer exceeds its capacity it first discards the oldest stored interaction and then stores the most recent one. Notably, the order of storage implicitly indicates the order of interaction, and also to which instance of the patient model it belongs. However, in practice DQN draws a uniform random sample of all stored interactions, thus removing not only correlations, but also sequential information.

For buffer composition variation during experiments, the buffers of the RL therapists were pre-filled with one of five datasets (see next paragraph) or remained initially empty. Each dataset consisted of interactions between an agent and a patient model, which were generated by exactly one pre-defined treatment protocol. The sixth buffer version remained initially empty without any pre-filled dataset. It filled up during training with each interaction that took place between the RL therapist and the patient model. Thus, the empty buffer was generated entirely through an RL-therapist’s own treatment policy, and the five pre-filled buffers resulted from divergent treatment policies enabled by DQNs’ off-policy property. The RL therapist can learn from these treatment

protocols to perform EL, while its own target and behavior policies may differ from the treatment protocols, which produced the interaction datasets.

Treatment Protocols. Following Bourgin et al. (2019), five synthetic datasets were created to introduce a prior into the RL therapist’s treatment policy for protocols that are typically used in lab experiments to perform EL. All datasets consisted of 100008 therapist and patient interactions from 4167 therapy sessions, with each session lasting 24 steps. In Tomasiak (2024) 24-step therapy sessions appeared to be particularly suitable to enable RL agents to learn EL. The default buffer size was 100000 interactions. To accommodate only complete sequences of RL therapist–patient model interactions, the buffer size was increased to 100008. This corresponds to 4167 complete 24-step therapy sessions ($24 \times 4167 = 100008$). The following five treatment protocols were used:

- **Immediate extinction:** The RL therapist shows exclusively cue A during stage 2.

For a long time it has been the default strategy in EL research. See Algorithm 2.

Algorithm 2 Immediate extinction protocol

```

for extinction trial  $t = 1, \dots, 24$  do
    perform action  $[1 \ 0 \ 0]$ 
end for

```

- **Random extinction:** The RL therapist selects actions completely at random. See Algorithm 3.

Algorithm 3 Random extinction protocol

```

for extinction trial  $t = 1, \dots, 24$  do
    choose random action from action space
end for

```

- **Fixed gradual extinction:** The RL therapist shows always cue A but never cue B. Additionally, cue A is paired with the US in the trials 1, 3, 6, 10, and 15. The fixed

pairings are equivalent to a gradually decreasing frequency of the US. It was adapted from Gershman et al. (2013) and resulted in successfully preventing the return of CRs in lab experiments. See Algorithm 4.

Algorithm 4 Fixed gradual extinction protocol

```

for extinction trial  $t = 1, \dots, 24$  do
    if  $t \in \{1, 3, 6, 10, 15\}$  then
        Perform action [1 0 1]
    else
        Perform action [1 0 0]
    end if
end for

```

- **Stochastic gradual extinction:** This protocol is similar to fixed gradual extinction. However, there are no fixed trials for pairings. The probability of the US being shown is gradually decreasing from 75% in trial 1 to 0% in trial 16. Thus, in the last 8 trials only cue A without the US is shown. See Algorithm 5.

Algorithm 5 Stochastic gradual extinction protocol

```

 $p([1\ 0\ 0]) = 0.25$ 
 $p([1\ 0\ 1]) = 0.75$ 
for extinction trial  $t = 1, \dots, 24$  do
    choose 1 of the actions according to probability
     $p([1\ 0\ 0]) = \text{Min}(p([1\ 0\ 0]) + (0.75/16), 1)$ 
     $p([1\ 0\ 1]) = \text{Max}(p([1\ 0\ 0]) - (0.75/16), 0)$ 
end for

```

- **Mixed extinction protocol:** The dataset consisted of an even mix of interactions from immediate extinction, random extinction, and fixed gradual extinction protocols across all patient models. See Algorithm 6.

Algorithm 6 Mixed extinction protocol

choose at random Algorithm [2](#), [3](#), or [4](#)

- **RL therapist’s own protocol:** Strictly speaking, this was not a protocol, nor was there a buffer pre-filled dataset. Interactions were generated by the RL therapist during training according to its own policy and then stored in the buffer.

Hyperparameters. DQN and its training process involve various hyperparameters. Hyperparameters used during Experiments are specified in Table [2](#). They were obtained by averaging over 20 optimization experiments with *Optuna*, each consisting of 50000 steps. That setup was computationally efficient, sufficient to observe convergence of the learning curve based on preliminary experiments, and was used in Tomasiak ([2024](#)). Slightly deviating from Tomasiak ([2024](#)), model evaluation was conducted after every 500 steps instead of 25000, by averaging the rewards over 100 patients or independent episodes instead of 1000, that were full RL therapist and patient interactions with their final non-zero rewards. Evaluation after 500 steps with 100 patients was chosen to be consistent with agent evaluation during the main experiments of the thesis, where finer-grained learning curves were achievable due to the availability of more data points. The *Optuna* experiments used the default tree-structured Parzen estimator algorithm, a Bayesian method, to narrow down the search space more efficiently. The objective was to obtain the highest reward in the final episode. However, it must be noted that due to an implementation error in the training pipeline, the learning rate used during Experiments 1-4 (see Section [5.1](#)) remained .0001, the default learning rate of DQN in *Stable-Baselines3*.

Table 2

Specification of hyperparameters used for DQN during experiments.

Hyperparameter	Value	Description from Raffin et al. (2021)
<code>tau</code>	.0545	It is the soft update coefficient (Polyak update $\in [0, 1]$). Default with <code>tau</code> = 1 leads to a hard update.
<code>gamma</code>	.8628	Discount factor Γ discounts future rewards. $\Gamma = 1$ values future and immediate rewards equally.
<code>learning_rate</code>	.0001	Learning rate, can be a function of the current progress remaining (<code>learning_rate</code> $\in [0, 1]$).
<code>batch_size</code>	897	Size of the mini-batch used for the gradient update.
<code>buffer_size</code>	100008	Size of the buffer, with stored interactions.
<code>train_freq</code>	6	Updates the model every <code>train_freq</code> steps.
<code>exploration_fraction</code>	.3076	Fraction of entire training period over which the exploration rate is reduced
<code>target_update_interval</code>	469	Update the target network every <code>target_update_interval</code> environment steps
<code>max_grad_norm</code>	9	The maximum value for the gradient clipping

Note. Except for the buffer size, and learning rate, the hyperparameters listed above were obtained by averaging values over 20 *Optuna* experiments, each consisting of 50000 steps and evaluation on 100 patients per logged reward. All hyperparameters not listed above used the default values of *Stable-Baselines3*.

5 Experiments and Results

5.1 Experiment 1 - Effect of Buffer Composition on EL with Off-Policy RL

The goal of Experiment [5.1](#) was to examine what effect the composition of the buffer had on EL performance of RL therapists that used the off-policy DQN algorithm. Therefore, six different models were trained to perform ET on the simple RW patients. The observation space in the environment was the 1 dimensional CR of the patient model. Each RL therapist was trained with a non-fixed buffer that was pre-filled at full capacity of 100008 steps with one of the following strategies: immediate, gradual deterministic, gradual stochastic, random, mixed extinction strategy, or an empty buffer. The standard non-fixed buffer stores new interactions and discards the oldest ones once its capacity is reached. To understand the choice of a 1 dimensional observation space and a non-fixed buffer instead of 2 dimensions and a fixed buffer see the experiments in the Appendix. All models were trained in parallel 10 times for 200000 steps.

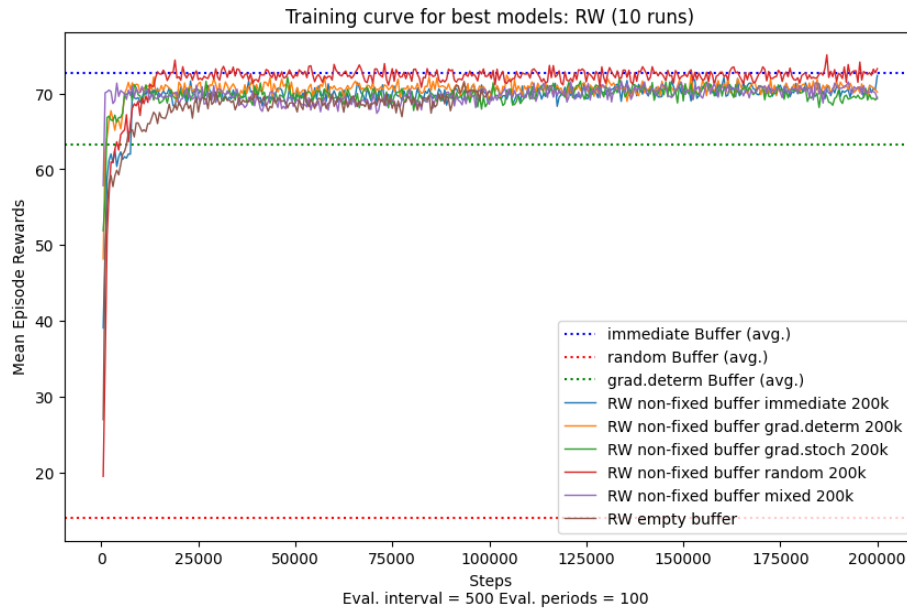
In Figure [2](#) Panel A shows the training curves for each model plotted in different colors, with the steps being shown on the x-axis, and the mean reward over 100 episodes and the ten runs of parallel training on the y-axis. The dotted horizontal lines show the mean reward achieved by manually applying the strategy of immediate, gradually deterministic, and random extinction. The dotted lines serve as a contextualizing baseline comparison of the rewards achieved by the RL therapist model to the rewards achieved by applying a predetermined EL strategy. Manually applying immediate extinction yielded the highest average reward, and random extinction the lowest. Contrary to that, the RL therapist model that used the buffer containing random actions achieved on average the highest reward compared to the other models, as early as after approximately 20000 steps. Besides, it reached average reward levels similar to the one achieved by applying immediate extinction manually. Likely, filling the buffer with random actions accelerated the learning through a vast amount of distinct experiences in the buffer, which in turn facilitated faster exploration of possible state-action pairs. Moreover, pre-filling the buffer with past

simulated random action data enabled by the off-policy property consistently outperformed the RL therapists with the initially empty buffer (trained only for 100000 steps) for all but the mixed extinction strategy. The pre-filled buffers served as the behavior policy, that contained state-action pairs produced by a diverging policy than the one that was the currently learned target policy.

In Panel B of Figure 2 the ten learned best policies of the RL therapist with random actions in their buffers were evaluated in more depth. It was checked what action the deterministic policy would choose upon observing a predefined set of 15 CR values, with $CR = \{1, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, 0.1, 0.05, 0.025, 0.0125, 0.00625, 0.003125\}$. A deterministic policy always chooses the action with the highest probability for the corresponding CR level or state mapping. Thus, the CR values were entered as observed environment states directly into the loaded models, without adhering to the previously used three stage procedure. The left hand-side of Figure 2 Panel B shows the policy of the best model that was found during training with RW patients using the random extinction buffer. The X-axis shows the run or the learned policy out of 10 trainings with a distinct model each, and the Y-axis shows the corresponding CR levels. The best action found corresponding to the CR level is color encoded, with 4 colors for 6 possible actions. Dots that are dark green indicate that the optimal action $[1\ 0\ 0]$ was found. Dots with lighter green indicate that either stimuli $[1\ 1\ 0]$ or $[0\ 1\ 0]$ were shown, without being paired with the US as can be seen by the zero at the third digit. Orange dots indicate that action $[0\ 1\ 1]$ was chosen, wherein cue B was paired with the US, but not cue A. Red dots indicate that either action $[1\ 0\ 1]$ or $[1\ 1\ 1]$ was chosen, wherein cue A was paired with the US leading to a strengthened association. On the right hand-side of Panel B the aggregate frequency of each action over all ten trained models can be seen, wherein each shown action has its own frequency and actions that were not performed by any model were omitted. The right hand side indicates that 92% of actions that were executed by the best found policies were using the optimal action of $[1\ 0\ 0]$ which is used by the immediate

extinction strategy. In addition, it shows that the red dots on the left hand-side can be attributed solely to action $[1\ 0\ 1]$ being shown in 2% of all actions, but not to $[1\ 1\ 1]$.

A



B

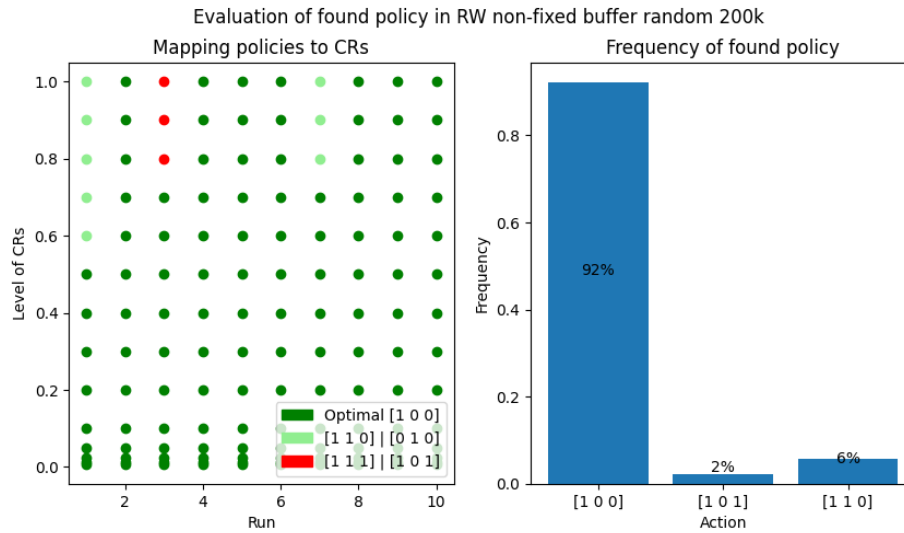


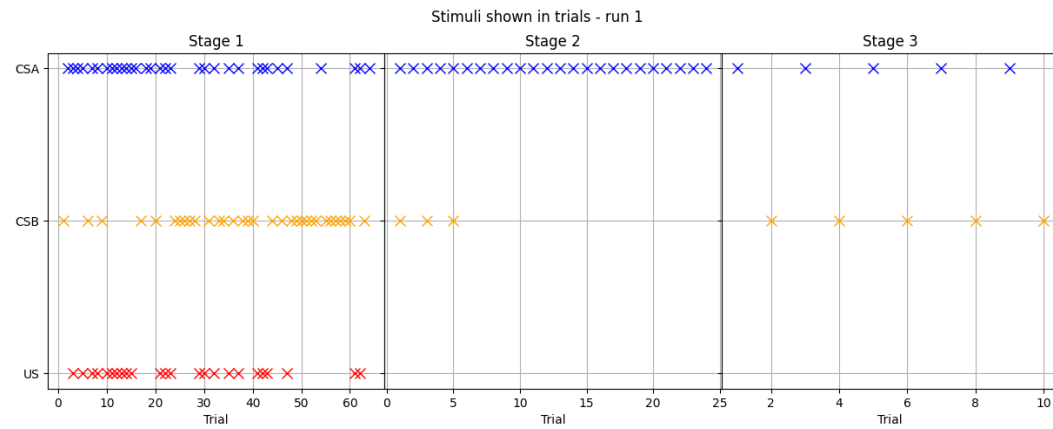
Figure 2

Training curves and policy evaluation of DQN RL therapists trained on RW patients

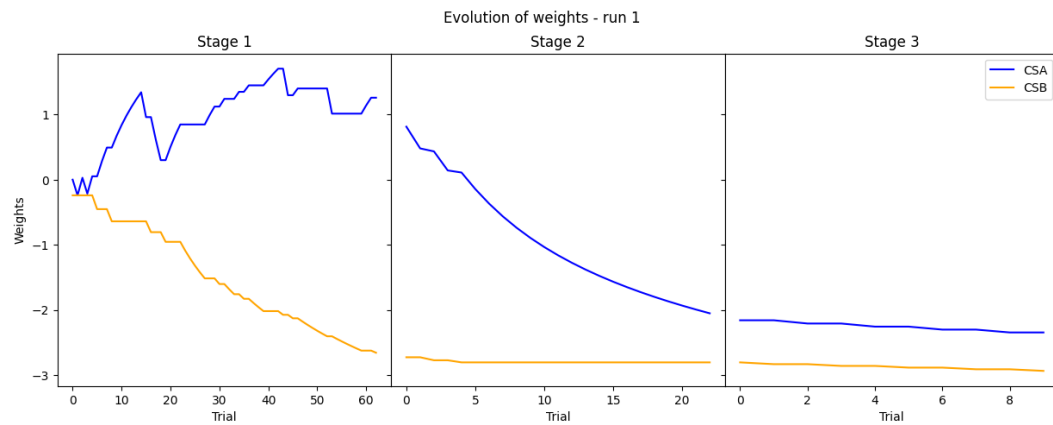
Note. Panel A: Comparison of achieved mean rewards during training of the DQN therapist models on RW patients. Each curve differs in respect to the data used to fill in the buffer. Random actions in the buffer outperformed the other conditions. Panel B: Evaluation of learned policies during training with a random buffer, where a run is an independently best trained model. Overall, the DQN model found the optimal behavior for 92% of the CR levels used for policy evaluation.

Figure 3 zooms into the performance and illustrates as an example a single episode or EL session performed by the best found policy in run 1. Panel A shows which stimuli were presented in each trial for all three stages. Panel B shows the evolution of the patient’s internal model weights for each stimuli over all three stages. Panel C shows the level of CR exhibited by the patient each trial after observing the stimuli. Figure 3 clearly indicates that the patient learned to exhibit a CR upon seeing cue A in stage 1. Moreover, it illustrates that the best model found in the first run, out of ten parallel runs, successfully alleviates the CR level of the patient by performing its learned policy. Lastly, a sharp decrease from the magnitude of the CR level at the end of stage 1 compared to stage 3 can be seen. Overall, the best trained models succeeded in performing EL and tended to use the optimal policy of $[1 \ 0 \ 0]$ during stage 2.

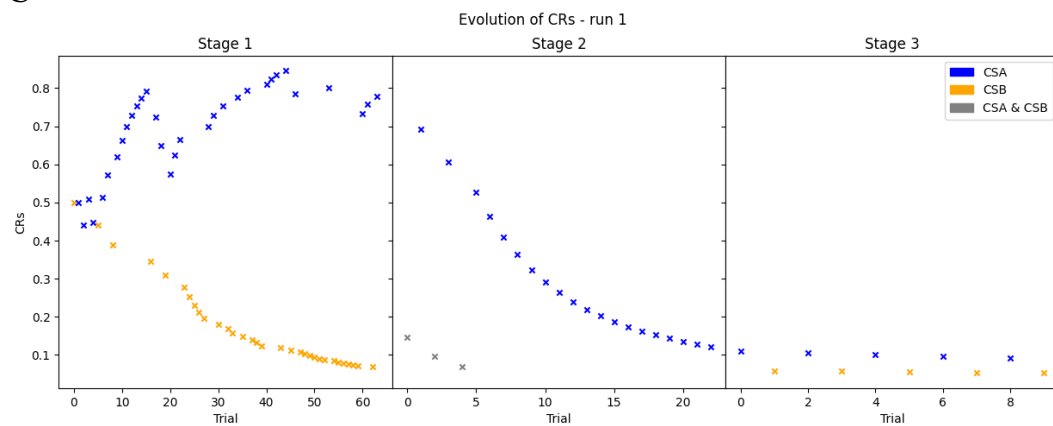
A



B



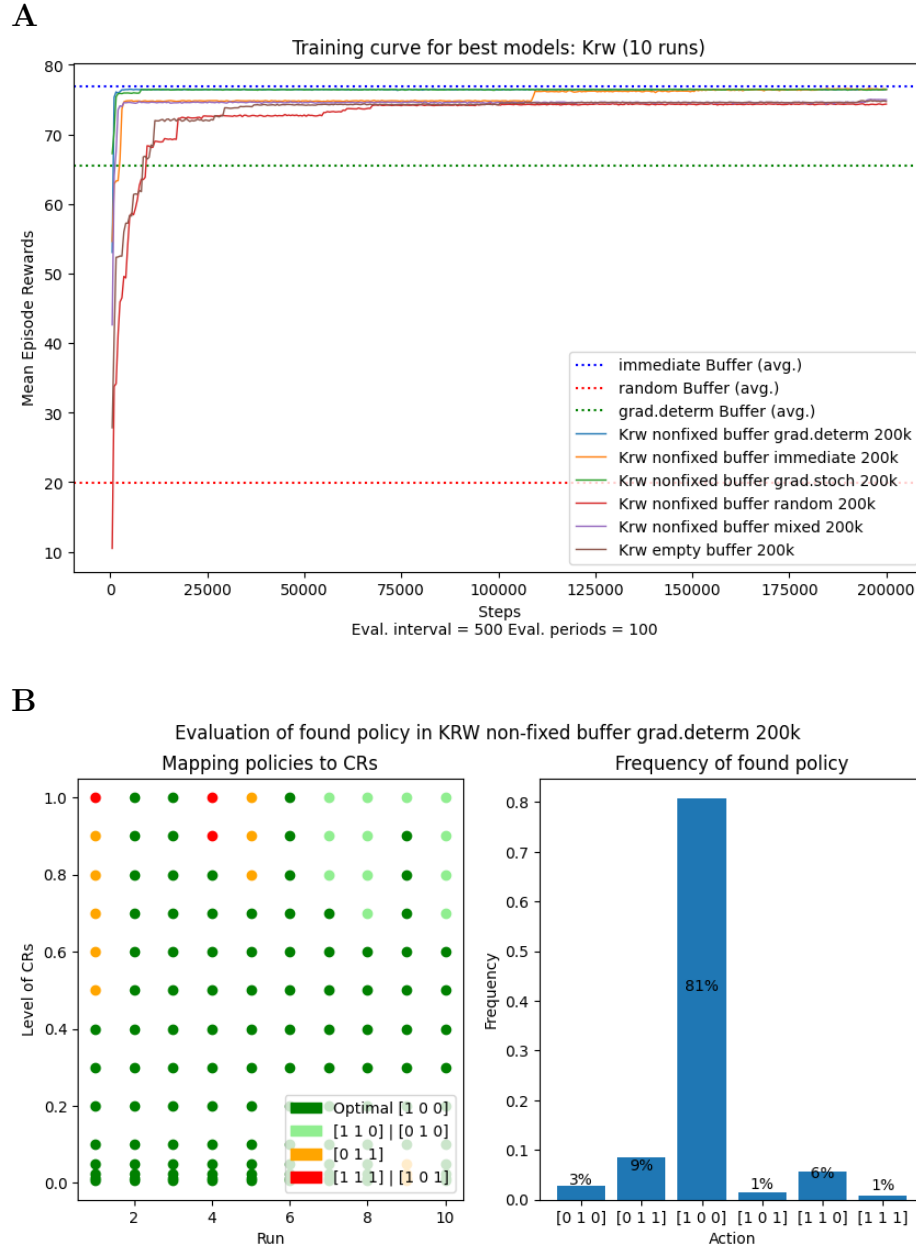
C

**Figure 3**

Successful EL - 3 stage interactions between the best model (run 1) evaluated in Figure 2 and an RW patient model during a single episode

Note. Panel A: Stimuli shown across all stages in experiment. Panel B: Evolution of the RW patient's weights regarding cue A and cue B across all stages. Panel C: Evolution of the observed RW patient's CR levels across all stages and trials.

Experiments with the same set up were performed for the KRW and DPKRW patient models, too. The KRW model is a stochastic generalization of RW and assumes a stochastic process for the patient’s association learning. In Figure 4 Panel A the training curves of RL therapists that interacted with KRW patient models are shown. Similarly, the average rewards of the predefined strategies that were applied on KRW patient models are plotted as dotted lines. A slightly higher average reward was achieved than in the RW condition. In the KRW condition the RL therapists that were trained with gradual extinction experiences in their buffer outperformed the others with on average higher rewards that were achieved earlier in training. Approximately, they reached the same reward level as the immediate extinction strategy baseline comparison, that achieved the highest average reward. Nevertheless, all trained models achieved on average similar rewards close to 85. In Panel B of Figure 4 the found policies of the best performing models that used the deterministic gradual extinction strategy were evaluated. In 81% of the test CR levels the most frequently performed action was $[1\ 0\ 0]$, suggesting it was the best action across CR states to perform EL on KRW patients. Similarly to the RW experiment, the RL therapists with the initially empty buffer learned slower and achieved on average lower rewards than the best performing buffer with immediate or deterministic graduate extinction data, although similarly high average rewards were achieved as with the random actions buffer.

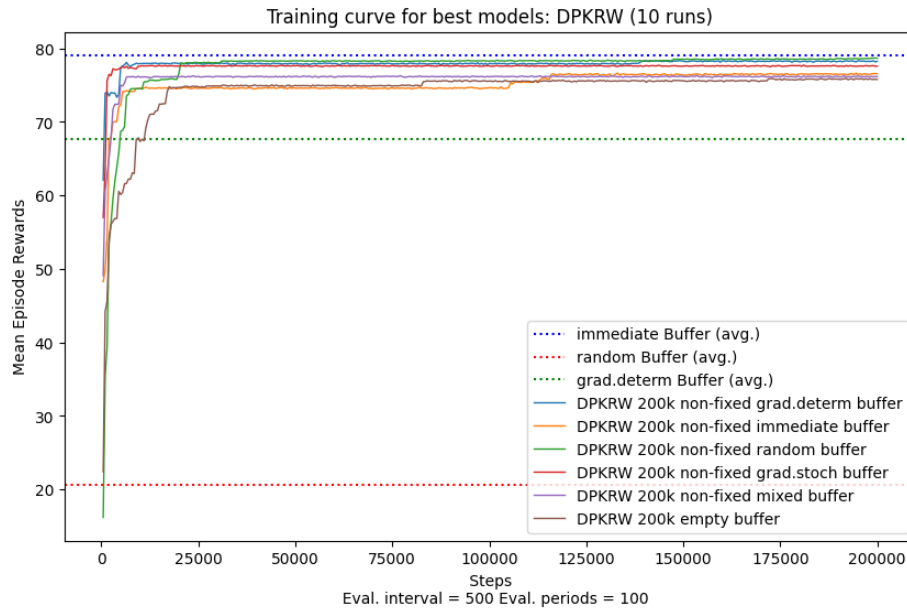
**Figure 4**

Training curves and policy evaluation of DQN RL therapists trained on KRW patients

Note. Panel A: Comparison of achieved mean rewards during training DQN therapists on KRW patients. Each curve differs in respect to the data used to fill in the buffer. The buffer with the gradually decreasing deterministic protocol outperformed the others. Panel B: Evaluation of learned policies during training with a gradually decreasing buffer, where a run is an independently best trained model. Overall, the DQN model found the optimal behavior for 81% of all CR levels used for policy evaluation.

Lastly, the DPKRW model extends KRW to allow patients to store multiple learned associations for different latent causes. In Figure 5 Panel A the training curves of RL therapists trained on DPKRW patients were plotted. The dotted baseline comparisons was the average reward achieved through predefined EL strategies performed on DPKRW patient models. Consistent with the RW patient condition, RL therapists trained with buffers that contained random actions achieved on average the highest rewards. The RL therapist trained with buffers that contained gradually decreasing deterministic strategy data and all other strategies performed similarly. All models achieved on average a similar reward close to the best baseline comparison strategy of immediate extinction with exclusively action $[1\ 0\ 0]$. In Panel B of Figure 5 the policies of the best models trained with random actions in the buffer were evaluated for each run. In 90% of the test CR levels the models performed action $[1\ 0\ 0]$. Consistent with previous experiments, the results suggested that action $[1\ 0\ 0]$ was the best known action to perform EL on DPKRW patients across CR states. Consistently, the RL therapist with the initially empty buffers achieved the lowest average rewards, although close to the rewards of the other pre-filled buffers. In summary, the RL therapists learned to perform EL by performing action $[1\ 0\ 0]$ across CR levels and across all three patient models. Pre-filling the buffer with data generated by simulations, where the patient interaction was determined by pre-defined extinction protocols, accelerated learning and achieved on average higher rewards than without pre-filling the buffer.

A



B

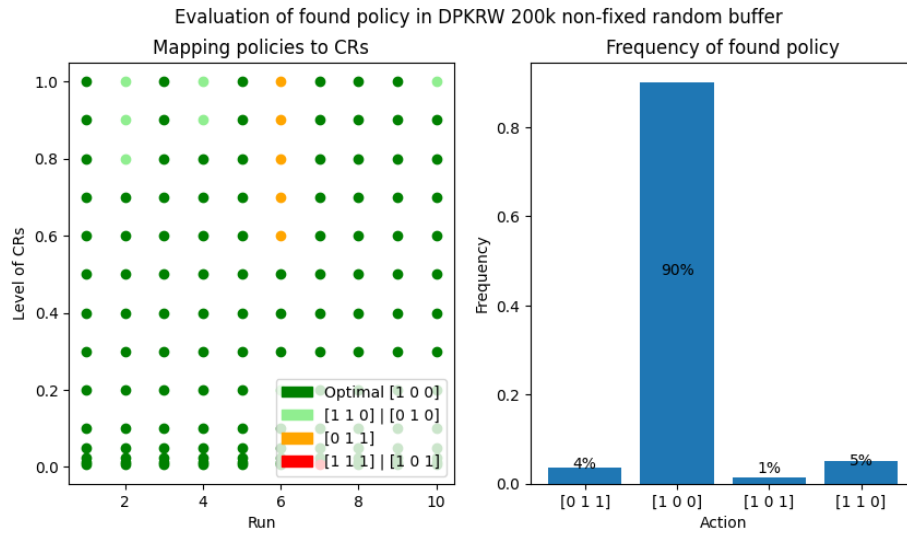


Figure 5

Training curves and policy evaluation of DQN RL therapists trained on DPKRW patients

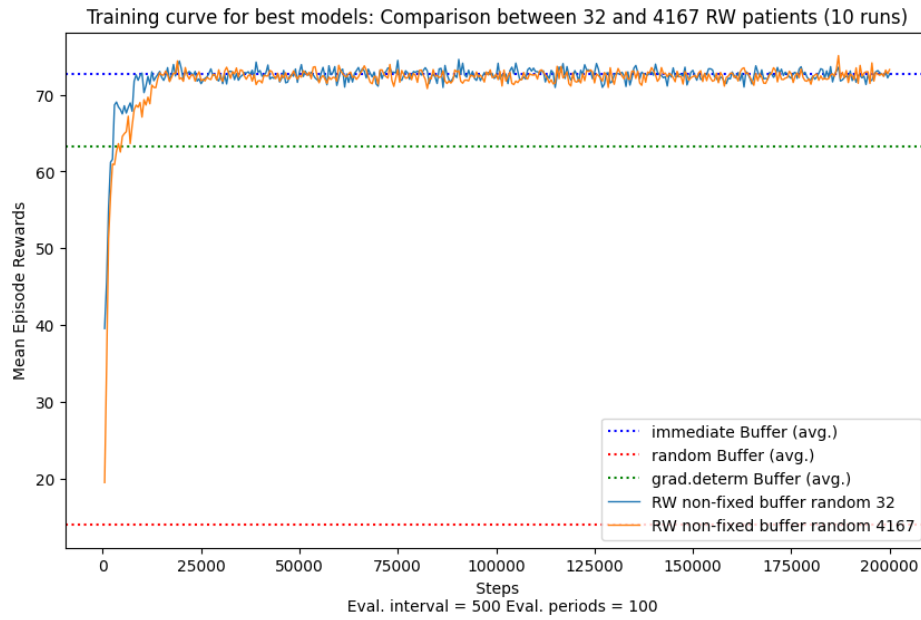
Note. Panel A: Comparison of achieved mean rewards during training DQN on DPKRW patients. Each curve differs in respect to the data used to fill in the buffer. The buffer with the gradually decreasing deterministic protocol outperformed the others. Panel B: Evaluation of learned policies during training with a gradually decreasing buffer, where a run is an independently best trained model. Overall, the DQN model found the optimal behavior or 90% of all CR levels used for policy evaluation.

5.2 Experiment 2 - Effect of Amount of Data in the Buffer on EL

The goal of Experiment 5.2 was to examine whether the buffer size had an impact on the RL therapists' trainings, or on their learned policies to perform EL on RW, KRW, and DPKRW patients. Initially, the used buffer size was 100008 which corresponded to 4167 patients. On one hand, such a high number of patients would be costly to collect for real patients. On the other hand, past patient data might impact training and provide more realistic patient behavior. Besides being costly it might be unnecessary to expose a high number of patients to any unintended risks of treatment or exposure, if the model can learn the optimal policy from a much lower number of patients. Thus, in Experiment 5.2 the buffer size of 100008 was contrasted with a buffer size of 768, which corresponded to 32 patients. Specifically, training was compared for models that used non-fixed buffers containing random actions. To allow for each simulated experience in the buffer being replaced by newer simulated experiences at least once, training steps were set to be 200000.

In Figure 6 Panel A it is shown that when training RL therapists on RW patient models both RL therapists with the respective buffer-size achieved similar rewards. In Panel B of Figure 5.2 it is evident that the model with 32 patients in the buffer mostly tended to find the optimal policy of $[1 \ 0 \ 0]$. However, for observed CR values close to 1 or 0 it sometimes exposed the patient to the US and paired it with cue A. The RL therapists with the smaller buffer size found the optimal policy for 89 % of the tested CR values, compared to 92% in the higher buffer size condition, and thus less frequently as the model with the higher buffer size (compare with Figure 2 Panel B). Hence, the buffer size had a slight impact on the performance of the model. However, using a smaller buffer size of 768 still allowed the RL therapist to learn a policy that mostly tended to perform EL with the optimal action of $[1 \ 0 \ 0]$.

A



B

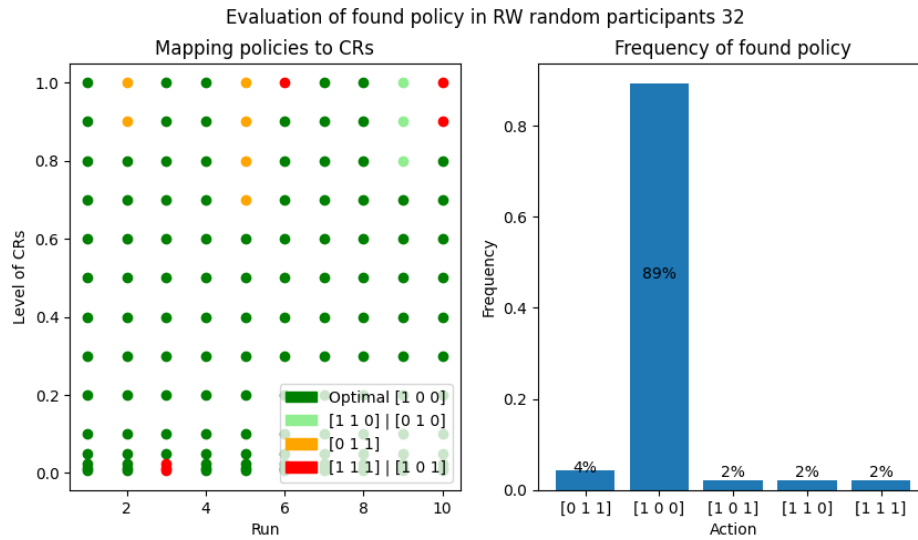


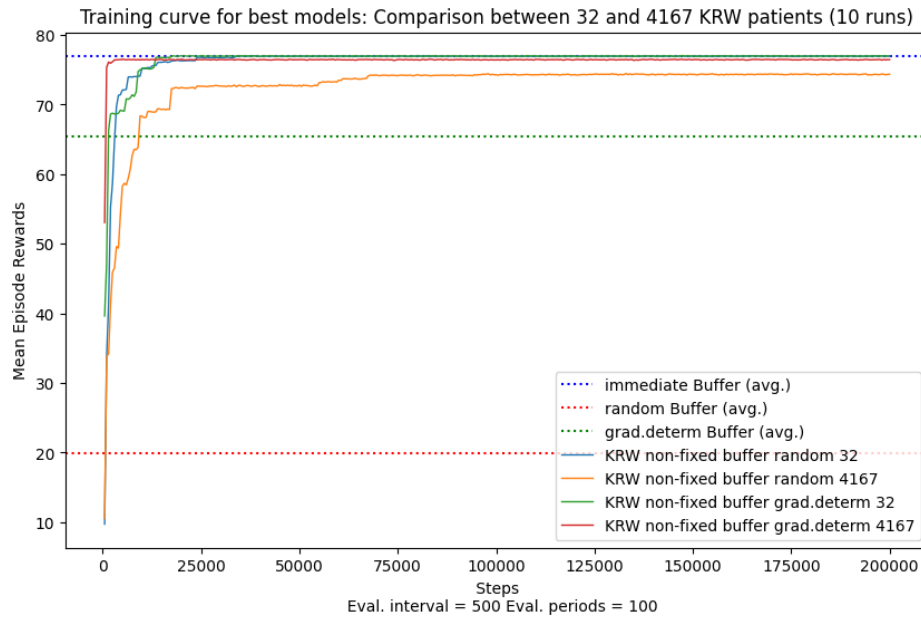
Figure 6

Comparison of training DQN RL therapists with different numbers of patient data in their buffers

Note. One patient equals 24 steps of stored interactions between the patient and the RL therapist. Panel A: Comparison of 32 patients in the buffer against the used default of 4167 patients with 100008 interactions. Panel B: Evaluation of learned policy with 32 patients in the buffer, where a run is an independently best trained model.

Similar results were demonstrated for RL therapists trained on the KRW, and DPKRW patients. For the KRW patient condition in addition to the buffers that contained random actions, buffers that contained experiences of the gradually decreasing deterministic extinction strategy were compared. This variation was introduced due to the results of Experiment 5.1 in Figure 4 Panel A, which showed that buffers with deterministic graduate extinction strategy achieved on average the highest average rewards and even faster than the other strategies. Still, the random extinction strategy was kept to ensure better comparability between patient models. In Figure 7 Panel A the four training curves of training with KRW patients and both buffer sizes are plotted. The RL therapists that used experiences of 32 patients with the gradually decreasing deterministic strategy achieved on average the highest reward and reached the average level of the immediate extinction baseline comparison. However, the RL therapists who used the buffer that contained data from 4167 patients with the same strategy achieved only a slightly lower average reward, but did so in less training steps. Consistently with experiment 5.1 the RL therapist using random actions with the default buffer size did not reach a similarly high average reward. Nevertheless, the RL therapists using the buffer with random actions from 32 patients did perform similarly to the RL therapists with the same number of patients in the buffer but a gradually decreasing deterministic strategy. In Panel B of Figure 7 the policies of the best performing models with buffers that contained data of 32 patients and that utilized the gradually decreasing deterministic strategy were evaluated. In 88% of test CR levels the models performed the best known action [1 0 0]. Moreover, it seems that in the rare cases that the models reinforced the association between cue A and the US it was for CR observation values close to zero.

A



B

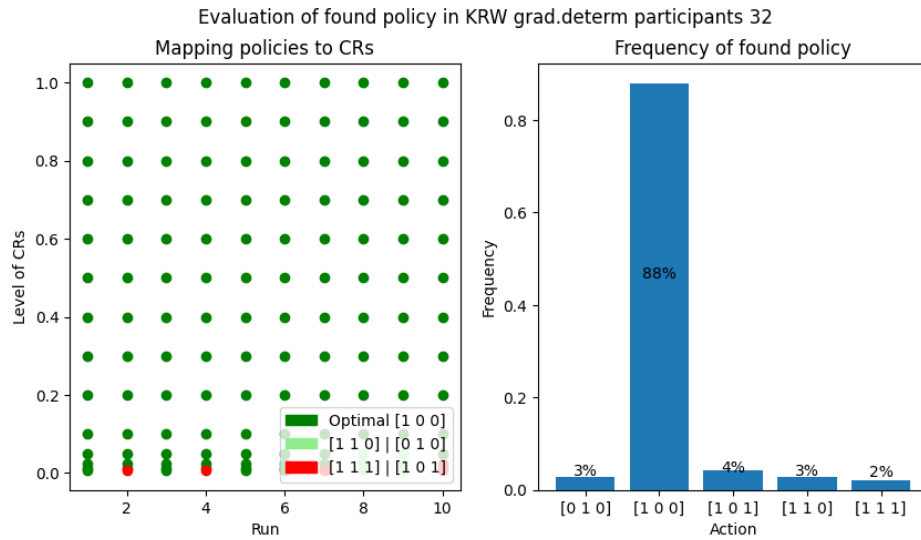


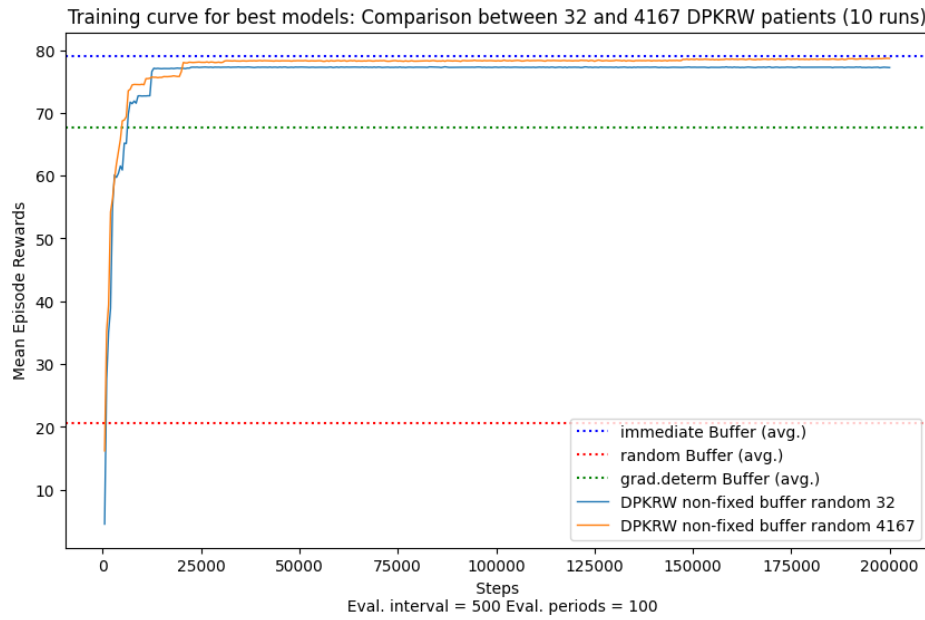
Figure 7

Comparison of training DQN RL therapists on KRW patients with different numbers of patient data in their buffers

Note. Trained on KRW patients. One patient equals 24 steps of stored interactions between the patient and the RL therapist. Panel A: Comparison of 32 patients in the buffer against the used default of 4167 patients with 100008 interactions. Panel B: Evaluation of learned policy with 32 patients in the buffer.

Lastly, RL therapists with different buffer sizes trained on DPKRW patients are plotted in Figure 8 Panel A. The highest average rewards were achieved by the RL therapists that utilized the data of 4167 patients in their buffer. Nevertheless, the RL therapists that used the smaller buffer size performed similarly. Both reached approximately the same reward level as achieved by applying the pred-defined immediate extinction strategy. To determine whether a smaller buffer size influenced the learned policies, we specifically assessed the performance of models trained with the smaller buffer size. In Panel B of Figure 8, the policies learned by the models trained with a smaller buffer size were evaluated. In 90 % of test CR levels the policies chose to perform the best known action of $[1 \ 0 \ 0]$. However, in 4% of test CR levels the policies chose to reinforce the cue A and US association by performing action $[1 \ 0 \ 1]$, slightly more often compared to 1% as seen in Figure 5 Panel B. Models trained with a larger buffer size performed slightly better, as can be seen in the evaluation in Panel B of Figure 5. Overall, the buffer size had only a negligible impact on the training and EL performance of the RL therapists across patient models.

A



B

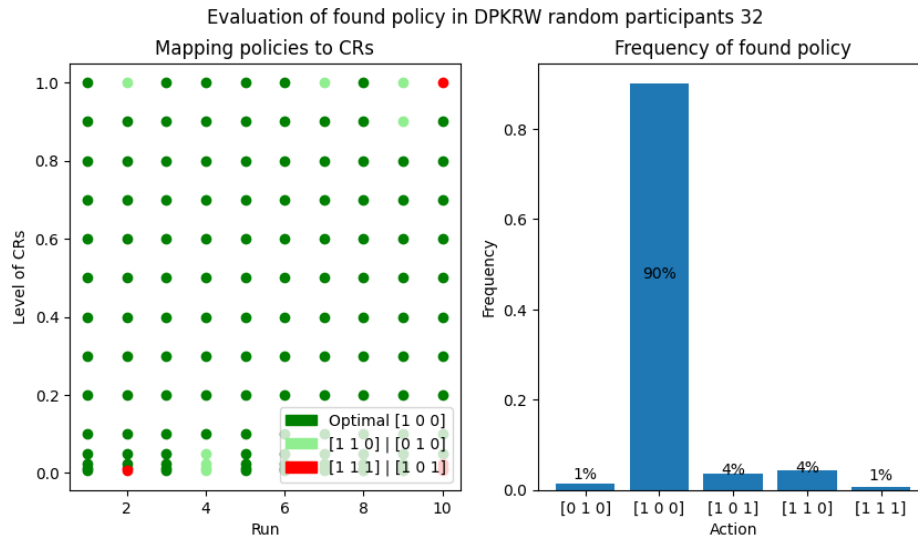


Figure 8

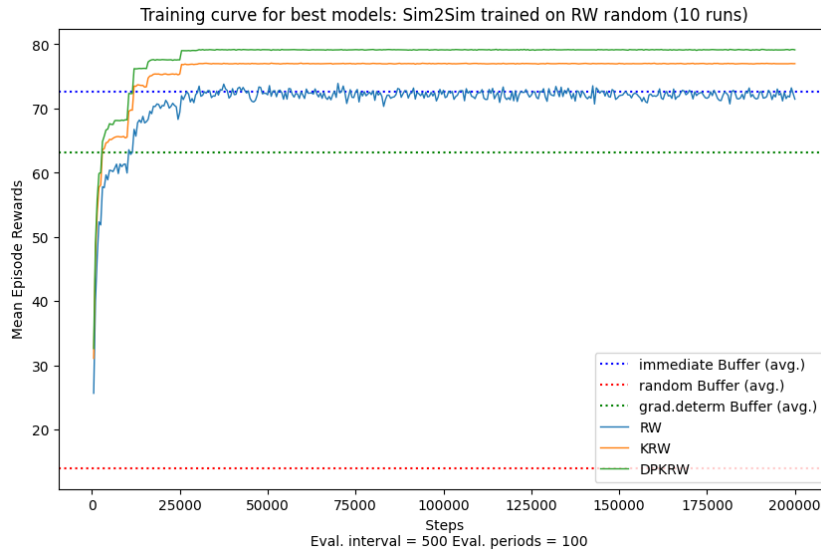
Comparison of training DQN RL therapists on DPKRW with different numbers of patient data in their buffers

Note. Trained on DPKRW patients. One patient equals 24 steps of stored interactions between the patient and the RL therapist. Panel A: Comparison of 32 patients in the buffer against the used default of 4167 patients with 100008 interactions. Panel B: Evaluation of learned policy with 32 patients in the buffer.

5.3 Experiment 3 - Effect of Differing Patient Models during Evaluation on EL Robustness

The goal of Experiment 5.3 was to evaluate whether RL therapists trained on one patient model performed similarly robust when tested on another patient model, a Sim2Sim evaluation. To assess the generalizability of the RL therapists' EL performance, it is necessary to examine whether the learned policy is overly specialized to the environment or data used during training and whether it also performs well on other patient models. To achieve this goal ten RL therapists were trained in parallel for 200000 steps. However, this time after every 500 steps the models were evaluated in regard to its reward three times. While previously the average reward of all ten runs was logged and the average of these was plotted, this time three different average rewards were logged for the same training step. One logged reward corresponded to the previous procedure and shows the performance when being trained on one patient model and then tested on the very same patient model. The two additional evaluations were conducted on the other two patient models that were not used during training. Notably, the ten trained models and their learned policies that determined which actions the RL therapists performed upon observing a CR remained the same. Thus, eventual differences in performance stemmed rather from a different patient model behavior. In Experiment 5.3 the only observed patient behavior was the CR and with it the implicitly inferred change in association strength. In Figure 9 Panel A the curves plotted resulted from training RL therapists on RW patients, and then logging the models' rewards when being tested on all three patient models. The dotted baseline comparisons were the average rewards achieved on RW patients by manually applying predefined strategies.

A



B

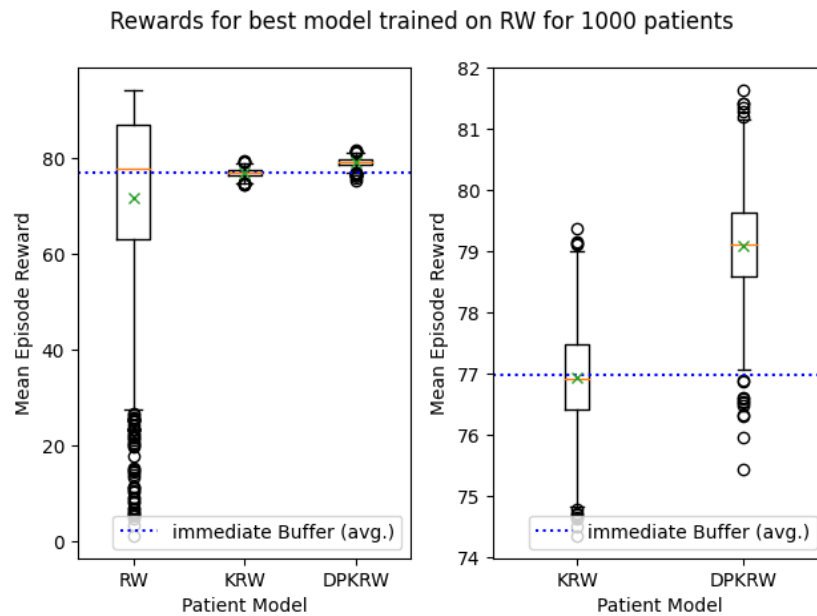


Figure 9

Sim2Sim: Training on RW and evaluation on RW, KRW, and DPKRW patients

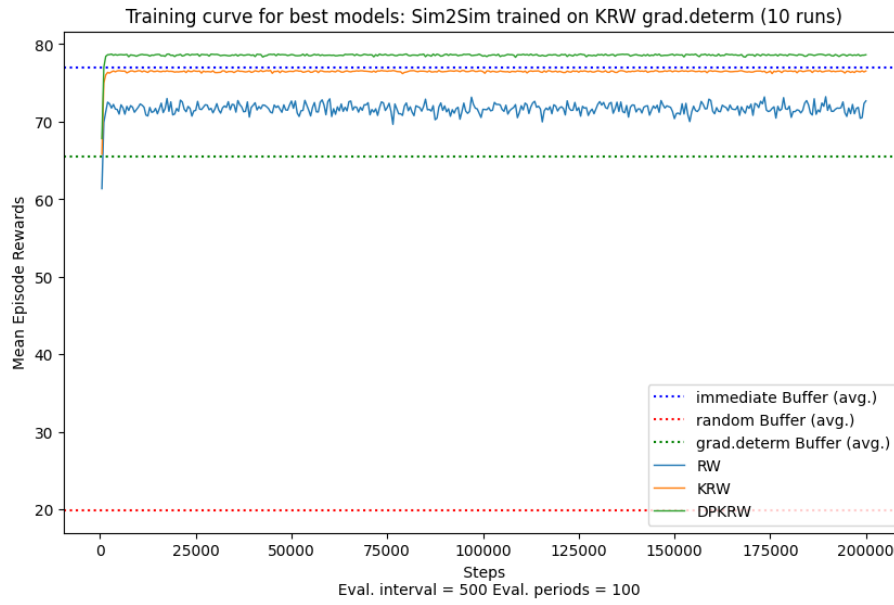
Note. Panel A: Training curves for DQN RL therapists trained on RW patients with a random buffer and evaluated on RW, KRW, and DPKRW patients. Panel B: Performance of final best model (run 1) trained on RW. Each box plot represents 1000 patients. Green crosses indicate the mean values. On the right hand side a zoom into the KRW and DPKRW box plots is shown.

The same models trained on RW patients achieved their highest average reward when being tested on DPKRW patients. The achieved average reward was even higher than the baseline comparison of applying immediate extinction on RW patients. Taking a closer look, in Panel B a box plot diagram shows the achieved average rewards of the same best policy of a single model. Each box plot shows the rewards obtained after 1000 patients, which was equivalent to the number of patients of ten parallel runs and an evaluation by averaging over 100 episodes or patients. The left hand side of Panel B shows that the orange lines which indicate the median values were almost on the same level when tested on all three patient models. However, the distribution of achieved rewards was wider and skewed when tested on RW patients, resulting in a lower mean reward shown by the green cross compared to the other two patient models. On the right hand side of Panel B the RW box plot was left out to obtain a zoom into the KRW and DPKRW models. Both box plots indicate a similar distribution of the achieved rewards. However, the mean and median reward values obtained by testing on DPKRW patients were slightly higher.

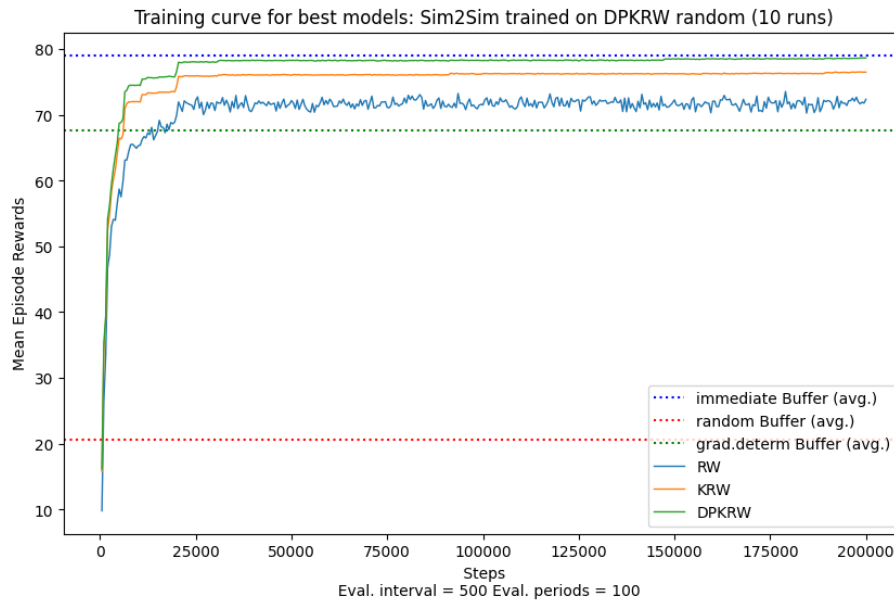
In Figure 10 Panel A three reward curves for RL therapists that were trained on KRW patients and tested on all three patient models with default parameters are shown. In Panel B the rewards of RL therapists trained on DPKRW patients and tested on all three patient models are shown. The baseline comparisons in each Sim2Sim plot were always the average rewards of predefined and manually applied EL on the same patient models as the ones the RL therapist were trained on in the respective figure. Consistently, among all three Sim2Sim figures when tested on DPKRW patients the highest average rewards were recorded, followed by rewards on KRW patients and then lastly on RW patients. Moreover, the rewards of the RL therapist that were trained on the same patient model as used in the baseline comparisons converged always towards the immediate extinction strategy. Therefore, the highest average rewards achieved when tested on DPKRW patients likely reflected patient model artifacts or patient parameter values rather than a better EL performance. The RL therapist remained the same and produced the

same actions for the respective CR levels across patient models. Although other parameter configurations might have changed the found best policy, the previously best found action of $[1\ 0\ 0]$ remained consistent across all three patient models. These results suggested that while all three models differed mathematically, they remained very similar in regard to the best known EL actions. Overall, in Experiment 5.3 the ET generalized well to other patient models than the ones that they were trained on. However, this finding might be limited by the relative similarity of all three patient models in regard to the best known actions found, and thus in future research models with more pronounced differences in patient behavior should be considered.

A



B

**Figure 10**

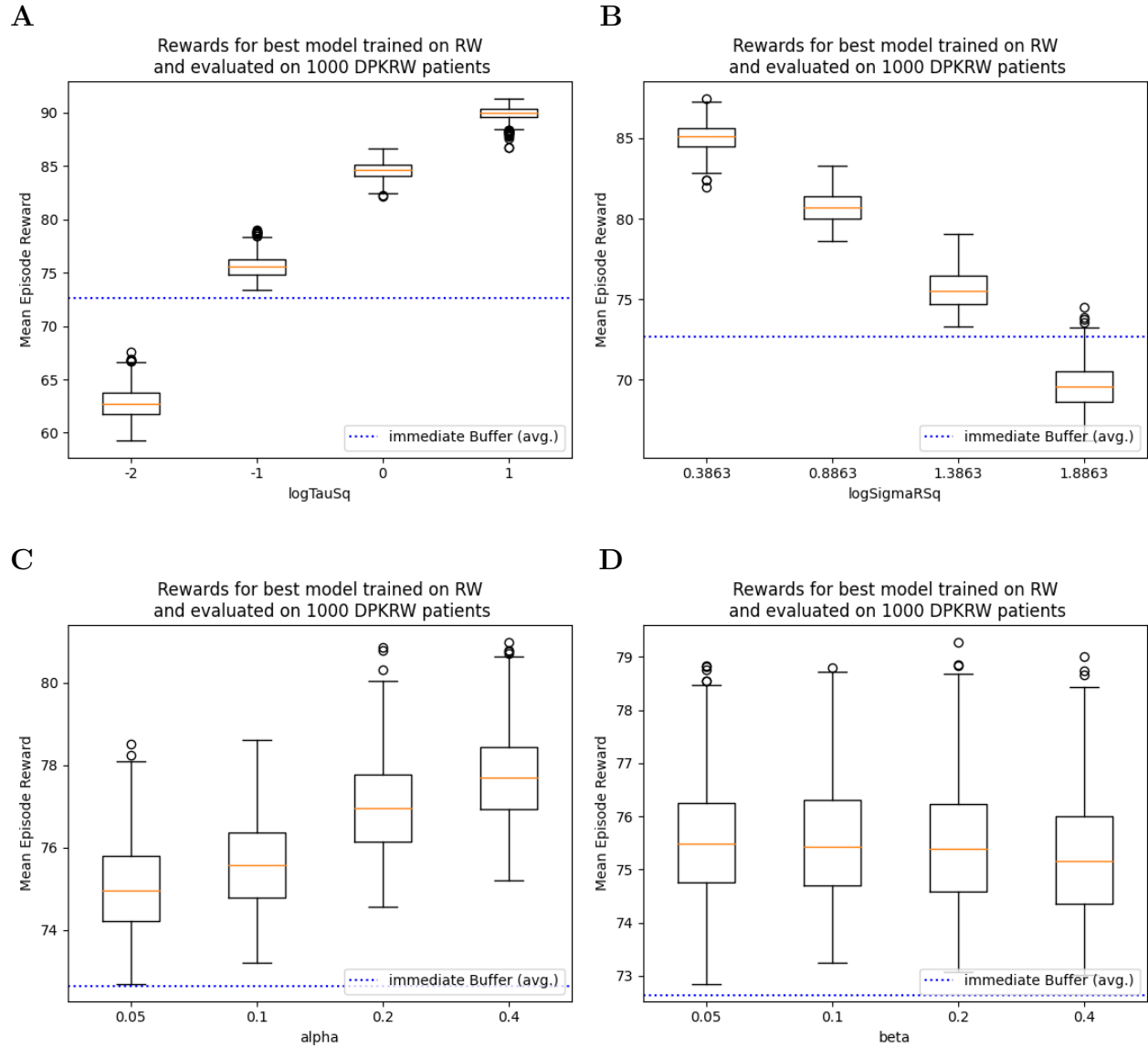
Simulation to Simulation: Training on KRW or DPKRW patients and evaluating on RW, KRW, DPKRW patients

Note. Panel A: Training curves for DQN trained on KRW patients with a gradually decreasing deterministic buffer and evaluated on RW, KRW, and DPKRW patients. Panel B: Performance of final best model trained on DPKRW patients with a random buffer. In both panels the same models achieve their highest rewards on DPKRW patients.

5.4 Experiment 4 - Effect of Shifting Parameters in Evaluation Patients on Robustness of EL Performances

In Experiment 4 the robustness of the RL therapists' performances was evaluated when the evaluation patient parameters differed from the training parameters. Differing parameter configurations in patient models might resemble different individual characteristics. For instance, patients with mood and anxiety disorders were observed to show reduced reward and increased punishment learning rates compared to control groups (Pike & Robinson, 2022). During all previous experiments the *alpha* learning rate was randomly sampled from a Dirichlet-distribution for each RW patient after each episode. That is, domain randomization was applied for RW patients during training and evaluation in Experiments 1-3. Thus, RL therapists learned to perform robust EL when applied on RW patients with differing learning rates α . RL therapists tended to perform action [1 0 0] independent of the patient models' learning rates α . In contrast, for KRW and DPKRW patients no domain randomization was applied during previous experiments.

In Figure 11, the best-performing model trained on RW patients was tested solely on DPKRW patients, while one patient parameter was varied across the panels and all other parameters were kept at their default values. In Panel A, each of the four box plots shows the rewards obtained from EL on 1000 DPKRW patients with four different $\log(\sigma_r^2)$ values, the log variance of state diffusion. In previous experiments, the default value of -1 was used and the evaluation was performed for 10 independent runs with 100 patients each. In Panel B, four different $\log(\sigma_r^2)$ values, representing the log variance of the reward, are plotted; the default value was 1.3863. In Panel C, the concentration parameter *alpha*, which influences the probability of inferring a new latent cause and had a default value of 0.1, was varied. Lastly, in Panel D, the stickiness parameter *beta* which influences the patients' probability of remaining at the currently inferred latent cause and had a default value of 0, was varied.

**Figure 11**

Performance differences on DPKRW patients for different parameter configurations

Note. Model was trained on RW with random buffer. Panel A: Performance difference for 4 different $\log(\sigma_\tau^2)$ values. Panel B: Performance difference for 4 different $\log(\sigma_r^2)$ values. Panel C: Performance difference for 4 different α_{DP} values. Panel D: Performance difference for 4 different β values.

The results suggest that $\log(\sigma_r^2)$ and *alpha* were positively associated with the average reward. In contrast, $\log(\sigma_r^2)$ was negatively associated, whereas *beta* did not appear to show any association under the tested parameter configuration. The lack of an apparent effect of varying *beta* may have resulted from the default α value, which may not have caused considerable changes in the inferred latent causes during all three stages. Nevertheless, across varied parameter settings, the learned policy remained unchanged and tended to apply action [1 0 0] across different CR values, indicating robust selection of the best-performing action under parameter shifts. However, this tendency may also indicate that the setup was too simple to fully assess the effect of shifting parameters on EL robustness, since realistic settings with differing individual characteristics might require individually adapted actions that vary across states. Likewise, while the domain randomization in KRW and DPKRW patients during training or model evaluation was not expected to have changed the results substantially, the lack of it further limits the generalizability of the findings.

6 Discussion

The goal of the present thesis was to evaluate whether off-policy RL therapists could learn to perform automated and robust EL on simulated patient models. In particular, it was examined how the buffer composition in off-policy RL impacted training, EL performance, and what policies were learned from past simulated data resembling different EL extinction strategies. Moreover, it was assessed whether being trained to perform EL on one patient model allows RL therapists to generalize their EL performance to other patient models, and to differing patient parameter configurations during evaluation compared to training. For these purposes 4 experiments were conducted in which the buffer composition of RL therapists that were trained DQN models, the buffer size, the patients simulative association model, evaluation patient models, and their parameter configurations were varied.

6.1 Q1 Training Off-Policy RL Therapists on Past Simulated EL Data

6.1.1 *Q1.1 The Effect of the RL Therapist’s Buffer Composition on EL*

Across all experiments the RL therapists learned to successfully perform EL on patient models. Using off-policy DQN models with pre-filled buffers that contained simulated data from past EL interactions accelerated the RL training and lead to on average higher rewards than without past simulated data. While pre-filling the buffer with random lead to better performance, this finding was confounded by the replacement of past simulated data by newer simulations along the training steps. Often the highest average rewards were achieved by the RL therapists as fast as after 10000 to 20000 training steps. While 20000 steps would correspond to approximately 800 study participants when conducted live, off-policy RL allows to re-use and learn from past patient data as demonstrated with the present simulations. The RL therapists’ achieved average rewards were compared to the average rewards achieved by applying three predefined baseline strategies. Across all experiments the RL therapists outperformed the baseline strategy of performing random actions, which produced the lowest average rewards. Furthermore, RL

therapists mostly tended to outperform the average rewards achieved by the strategy of gradual extinction. That strategy achieved the second highest average rewards and is considered promising in preventing the relapse of CRs in the the aftermath of ET (Gershman et al., 2013; Shiban et al., 2015). Lastly, the RL therapists mostly reached a similar average level of rewards as reached by applying the strategy of immediate extinction, which achieved the highest average rewards. The two later of the three strategies are established procedures in EL experiments. Hence, in terms of reward the RL therapists with pre-filled buffers reached a comparable performance in the simulations to the immediate extinction strategy, that is frequently used in EL experiments.

Importantly, the RL therapists tended to learn meaningful policies that produced sensible actions during EL. Among RL therapists the best learned policies were mostly to perform the strategy of immediate extinction. For a vast majority of observed CRs the RL therapists tended to perform the action $[1\ 0\ 0]$, which when applied exclusively corresponds to immediate extinction. While immediate extinction is a widespread strategy in EL experiments it recently came under suspicion of being involved in causing relapses of CRs (Gershman et al., 2015), effectively counteracting the ET goal of long term unlearning of CRs. In contrast, the strategy of gradually introducing extinction was found to better prevent the relapse of CRs (Gershman et al., 2013; Shiban et al., 2015) and is thus particularly promising for ET practitioners and RL therapists. Nevertheless, in the present simulations the best known strategy when applied manually with a predefined protocol, was indeed immediate extinction, as seen by its consistently highest average rewards at the baseline comparison level across all three RW, KRW, and DPKRW patient models. Therefore, the RL therapists learned to imitate the best available strategy to perform ET in the environment and learning the strategy of gradual extinction instead would have been sub-optimal. However, since the strategy of gradual extinction is suspected to better prevent relapses of CRs in real patients, future research could focus on how to train the RL therapists to perform that particular strategy in order to achieve better automated ET

with real patients. In summary, the RL therapists learned to successfully perform EL with sensible actions on simulated patients and the buffer composition impacted the training speed and EL performance, thus demonstrating the benefit of using off-policy algorithms.

6.1.2 Q1.2 Amount of Data Necessary for RL Therapists to Learn EL

RL therapists using data from past simulated EL interactions during their training improved their EL performance faster and achieved on average higher rewards than those that used initially empty buffers. Moreover, it became evident that the data of 32 patient model’s EL sessions, each incorporating 24 patient therapist interactions, was sufficient to reach a performance level similar to the default RL therapist versions having data of 4167 patient model EL sessions. Fewer data points were not explored, as future real-world data collection would require a minimal level of diversity and sample size to ensure patient representativeness. A potentially promising approach could involve first training RL therapists with simulated patient models to enable full state-action pair exploration, then fine-tuning and evaluating them by using real patient data in the buffer. This would produce RL therapists better adapted to real human behavior, enhance safety, and potentially increase patient trust. Despite relying solely on artificial patient data, RL therapists trained on a 32 patient dataset still achieved performance comparable to the immediate extinction EL strategy, and required only a modest amount of patient training data in the buffer.

6.2 Q2 Robustness of EL Performance by RL Therapists

6.2.1 Q2.1 Robustness of EL Performance on Models Differing from the One Used for Training

Overall the RL therapists achieved a robust performance with consistently high average rewards. Moreover, the performance remained robust when applying EL on the other two patient models than being trained on. This was true across all the three patient models RW, KRW, and DPKRW. Although the patient models differed conceptually, mathematically, and in their complexity or number of parameters, using them for the RL

therapist training produced remarkably similar actions for unlearning the CR. The nearly constant achieved average reward differences across patient parameter configurations and across the applied baseline comparison treatment protocols, despite policies tending to produce similar actions, suggest that differences may be rather attributed to patient model artifacts than to a better EL performance learned by the RL therapists. Moreover, if performing $[1 \ 0 \ 0]$ always was the best action, a rule based approach might be more efficient than training an adaptive RL therapist. In contrast, while immediate extinction is a well established EL strategy, many ET manuals highlight the need of adaptiveness to individual patients or contexts (e.g., Abramowitz et al., 2019; Smits et al., 2019; Telch et al., 2014), and thus the RL therapists might have learned unrealistic therapist behavior that could be sub-optimal or non-robust in real world application. Therefore, it remained unclear whether the RL therapists would perform robustly on patient models that are more pronouncedly different or more empirically realistic and would require a higher degree of adapting the RL therapists' actions.

6.2.2 Q2.2 Robustness of EL Performance on Different Parameter Configurations of Patient Models

The RL therapists also performed robustly within a patient model. By randomly sampling the RW patient model's learning rate α and trying out different parameter configurations for the other two patient models, which can be seen as proxies for inter-personal differences in learning behaviors, a robust ET performance was demonstrated within a patient model in addition. Nevertheless, the findings on the effect of shifting patient parameters are constrained by the missing domain randomization for the KRW and DPKRW patient models. On the path towards automated ET a robust performance within a patient model for different patient configurations is as important as a robust performance across different patient models. Decisively, for both within patient and between patient robustness, the patient models must reflect empirical behavior to enable the RL therapist to learn realistic patient behavior and the appropriate responses to it. It

will be essential to test the robustness of the RL therapist’s performance on additional models as well as on real patients, before automated ET can be applied safely on a broad scale in clinical settings. In sum, in this thesis the RL therapists performed ET robustly within and between patient models. Nonetheless, precaution must be exercised since the patient models used for evaluating the performance robustness might have not been sufficiently distinct, in particular regarding to the best found actions.

6.3 Limitations and Future Outlook

Notably, the findings of this thesis are limited by its simulation approach that does not consider real patient data, the similarity and simplicity of the patient models in use, the restricted observation and action space available to the RL therapists, and an evaluative approach that may not have accurately captured long-term effects of unlearning the CR. First, the RL therapists in this thesis were trained on simulated data, and performed ET on simulated patients. While the present simulations were inspired by real world EL experiments, it is possible that crucial aspects of performing EL or unlearning CRs were not comprehensively reflected, as no real EL, no real therapist and patient interaction, and no real patient unlearning CR data were used. Introducing real patient data in each of these components of future experiments, might improve the simulation, increase the external validity of its findings, and increase confidence in applying RL therapist in real world EL or clinical settings. Potentially, RL therapist could be trained with patient models to perform ET well, and then fine-tuned with real patient data to obtain an RL therapist that is more adaptive to realistic human behavior instead of artificial patient model data. Such an approach would not only consider a more realistic data sample, but could be considered more safe and increase the trust of potential patients, since human data was involved in training and evaluating the RL therapists.

Second, while the patient models differed conceptually and mathematically, for instance by introducing a probabilistic learning assumption in KRW, or by adding multiple associations in parallel for the same stimuli combination in DPKRW, it was demonstrated

that effectively they remained similar in regard to the best found action to unlearn a CR. The similarity of the patient models limited the generalizability of the RL therapists' EL performance and their EL robustness on other more distinguishable patient models might be impaired. Besides the similarity of the patients models, due to simplicity, the experimental design allowed only for learning CRs in a two-stimulus setting. While such a design with only few stimuli is common in CC experiments for EL, it might have not captured the complexity of the mechanisms involved in clinical settings like panic, phobias and anxiety, stress, or obsessive-compulsive disorders. Thus future experiments, could consider more distinguishable or more complex models to strengthen the applicability of their findings to clinical settings.

Third, the findings on the robustness of EL performance are partially limited by the lack of within-model parameter shifts. In other words, no domain randomization was applied in the KRW and DPKRW patient models, neither during training nor during evaluation. A random parameter shift in the patient model could have better reflected inter-individual differences while still assuming the same underlying patient model. Nevertheless, based on the domain randomization used during RW patient model evaluation and the strong similarity in the best policies found across all three patient models, it is expected that parameter shifts would not have substantially affected EL robustness. In addition, the technique of weight variation between single rewards (Roggeveen et al., 2024) was not applicable here, since only the terminal reward was non-zero. Evaluating RL therapist performance on the EpiCare simulation benchmark for longitudinal clinical treatment regimes (Hargrave et al., 2024) could have provided further evidence for robust EL performance. However, this was not investigated in the present thesis. Thus, the lack of domain randomization for KRW and DPKRW patients, and the missing EpiCare benchmark evaluation constrain the present findings on EL robustness.

Fourth, the observational and action space of the RL therapist may not have been sufficiently comprehensive. In the present simulations, the RL therapist relied on a

one-dimensional observation space, limited to patients' CR. However, in practice, the data available to the RL therapist should be at least as comprehensive as the information used by clinical practitioners when forming treatment decisions or selecting stimuli to present during ET (Gottesman et al., 2018, 2019). For instance, clinical practitioners would consider the socioeconomic status, comorbidities of the patient, concurrently used coping strategies, or other health risks of the patient before forming their treatment decisions. Thus, a multi-dimensional observation space should be preferred to allow the RL therapists to infer meaningful decisions considering all relevant data points. Likewise, in the present simulation, the decision was limited to binary choices, whether to select a stimulus or not. Yet, in practice, there is typically a broader range of actions available to induce an association. For example, the association can be modulated by presenting only stimuli that are similar to the original trigger of the CR, altering the duration of exposure, or by concurrently administering medications, which is usually an option for treating patients that exhibit severe disorder. Similarly, a CS used in VRET could be tailored to the patient's therapy needs (Mahmoudi-Nejad et al., 2024). Hence, future experiments could consider a broader set of actions, that more realistically correspond to the actions undertaken by clinical practitioners.

Fifth and last, the evaluation whether the patient model exhibited a CR after EL did not accurately reflect long-term effects. The goal of ET is to decrease the CR or clinical condition on a long-term or erase it entirely. Empirically, the CR reappeared, or could be reinstated or renewed after a seemingly successful ET (Haaker et al., 2014; Quirk, 2002; Vervliet et al., 2013). Thus, it is essential that the success of automated ET reflects long-term effects in the evaluation. However, in the present simulation the passing of time was modeled by a step or an action performed by the RL therapist. Long-term effects were measured as the subsequent ten steps after performing EL. Modeling time this way can hardly be compared to the human experience or to the CR evolution after months. Furthermore, next to other variables the long-term effect of ET should be accurately

reflected in the reward function to allow the RL therapists to learn a meaningful actions (Gottesman et al., 2019). However, without a more complex modeling of time steps it is not clear whether long-term effects were sufficiently reflected in the used reward function. For instance, the learned policy was to perform immediate extinction during EL, a strategy which was suspected to be involved in the relapse of CRs (Gershman et al., 2013; Shiban et al., 2015). Thus, the learned policy may have indicated that long-term effects might have been neglected by the RL therapists or were not properly modeled in the simulation. Future experiments, may find ways to better reflect long-term effects in their simulations

7 Conclusion

In this thesis, the potential of automating ET through RL was demonstrated. More precisely, it was shown that RL therapists feasibly benefit from using past EL data, that stemmed from RL therapist interactions with simulated patient models during EL experiments, that are lab models of ET. Moreover, a robust EL performance was demonstrated within and between three patient models. Nevertheless, the findings were limited by the thesis’s simulative approach, that lacked real patient data, used three distinct but similar patient models, considered for the RL therapists only a restricted observation and action space, and was evaluated in a manner which does not compare to human long-term therapy success after months. To advance the automation of ET future studies will need to incorporate real patient data, use a broader set of patient models, and train RL therapists that consider more clinically relevant variables in their treatment decisions. To increase trust among potential patients and clinical practitioners and eventually pave the way for automated ET, it will be essential to evaluate how the RL therapists perform ET on real patients in clinical settings.

Appendix

Abbreviations

Table A1

List of used abbreviations in alphabetical order

Abbreviation	Meaning
CBT	Cognitive Behavioral Therapy
CC	Classical Conditioning
CR	Conditioned Response
CS	Conditioned Stimulus
DPKRW	Dirichlet-process-Kalman-Rescorla-Wagner model
DQN	Deep-Q-network
EL	Extinction Learning
ET	Exposure Therapy
KRW	Kalman-Rescorla-Wagner
LCM	Latent-Cause Model
LDS	Linear-Gaussian Dynamical System
MDP	Markov Decision Process
NS	Neutral Stimulus
POMDP	Partially observable Markov decision process
RL	Reinforcement Learning
RW	Rescorla-Wagner
Sim2Sim	Simulation to Simulation Test
UR	Unconditioned Response
US	Unconditioned Stimulus
VRET	Virtual Reality Exposure Therapy

Preliminary Experiments

Experiment A1

The goal of Experiment [A1](#) was to see whether an adaptation of the buffer could improve ET performance and whether it was beneficial to finding the optimal policy earlier or increasing its frequency when conducting ET on RW patients. Hence, Experiment [A1](#) was conducted with a similar setup like the experiments in the Experiment and Results section , but with the major difference that an adapted version of the buffer was introduced. Previously, when the RL therapist performed an action during training, the action, its associated CR level, and its achieved reward were stored in the buffer. In case that the buffer had been at full capacity already, the longest stored entry was deleted and the most recently performed entry was saved instead. While, this way the DQN algorithm is capable of adapting to a changing environment and learning from previous experiences, it results in the replacement of all experiences in the buffer after training the model over sufficient steps. In experiment [A1](#) the buffer was adapted into a fixed buffer to keep stored experiences and not to replace them after a new action. The adaptation ensures control over the actions in the buffer and allows learning from real world patient data without replacing it by new experiences during training on a simulated environment with a patient model.

Figure [A1](#) Panel A shows the training curves for DQN models trained on RW patients. There are three curves that resulted from training with an fixed buffer, and one from an empty buffer without any adaptations. The first three resulted from one buffer containing immediate extinction, the second one gradually fixed extinction, and the third one random action extinction experiences. Since, the stochastic extinction strategy was not entirely distinct in its conception and its results in the Experiments and Results section compared to a gradually fixed extinction, it was omitted to save computational time. The same was true for the mixed extinction strategy. Moreover, for the same reason the training steps were reduced to 100000. Again, the model resulting from a random action

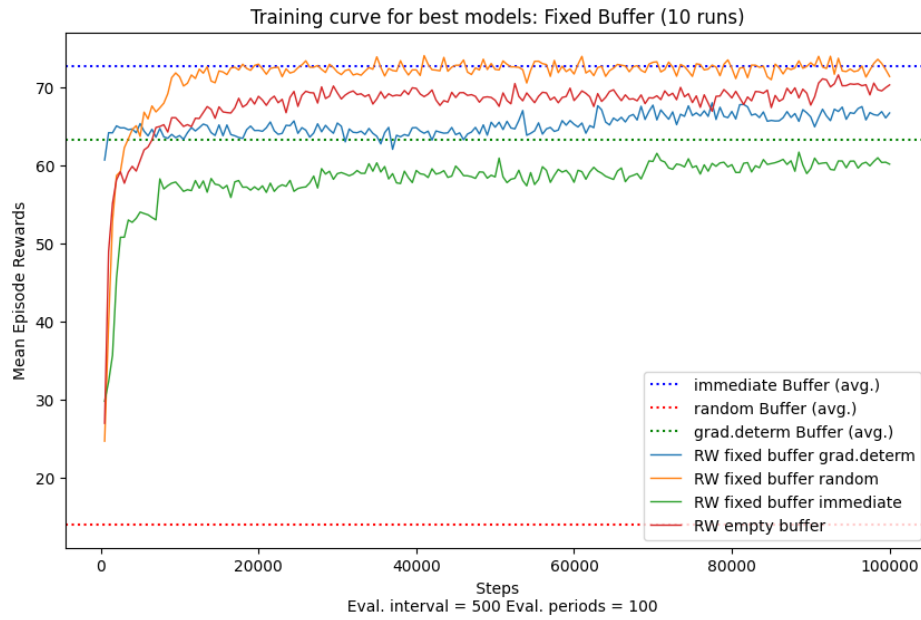
buffer achieved on average the highest rewards.

Deviating from Experiment 1 in the Experiments and results section, the other training curves performed consistently worse and did not converge to similar average reward levels. Likely, the adaptation of the buffer caused a slow down in learning and exploration since no new experiences are stored and the learning update is performed from a buffer sample. While sufficiently different experiences exist to allow exploration in the random buffer, in the other buffers the policy is fitted to only a subset of potentially possible action state pairs. However, examining the policies in more detail, the higher average reward appears to be misleading. In Panel A of Figure [A2](#), it is evident that one of trained models for the immediate fixed buffer appeared to be an outlier, that frequently showed action $[1\ 0\ 1]$. On the other hand, seven out of ten models, that were trained with an random fixed buffer learned to perform $[1\ 0\ 1]$ and $[1\ 1\ 1]$ for low CRs. Thus, it is inconclusive which fixed buffer lead to better result when trained for 100000 steps. In Panel B of Figure [A1](#) a training curve comparison between the best performing models, that used a random fixed or non-fixed buffer were plotted. In addition, the inter-quartile range (IQR) is shown as the light red or light green area. The Panel B of Figure [A1](#) suggests that the performances of both models, with fixed and non-fixed, were almost indistinguishable in regard to average rewards and therefore it had no apparent benefit to adapt the buffer for training with simulated data. However, when evaluating the RL therapist with real world data a fixed buffer will be necessary due to a limited amount of available data. Experiment [A1](#) demonstrated the feasibility of training with a fixed buffer.

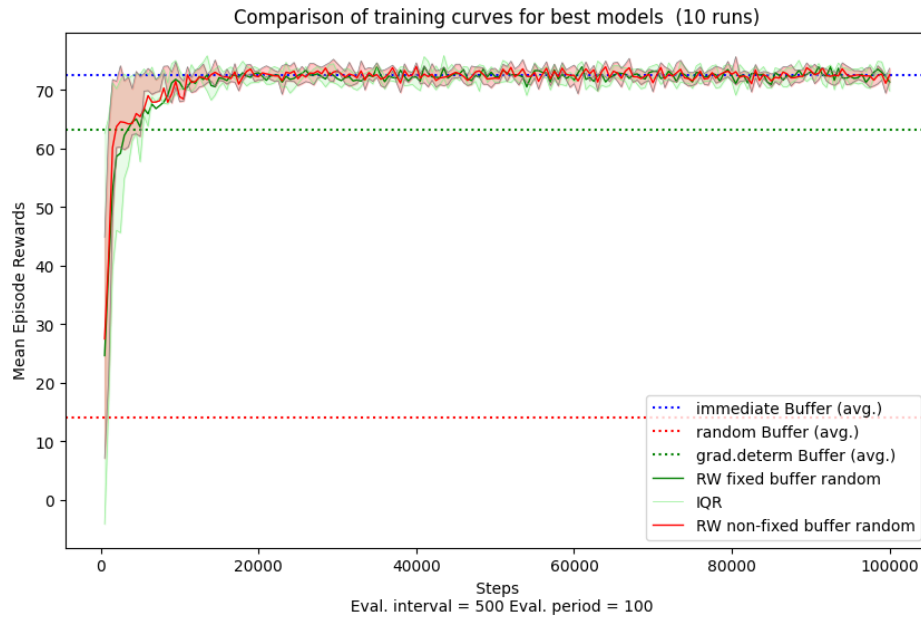
Furthermore, in Figure [A3](#) the evolution of the DQN models' Q-functions is plotted as an example for the first training run out of ten for each condition . The Q-function is plotted for three CR observation test values, a high $CR = 0.5$, a middle $CR = 0.2$, and a low $CR = 0.005$. Each curve shows the evolution of Q-Values for one particular action. In Panel A the Q-function values of a model trained on RW patients with an fixed buffer containing immediate extinction experiences stored in the buffer were plotted. The

immediate extinction buffer contained only action $[1\ 0\ 0]$. It can be seen that the Q-values for that action had much higher variability over the training steps compared to all other actions that were not in the buffer. Likely, using a deterministic policy with no action variance in the buffer data had an adverse impact on the exploration of other actions, and thus the Q-values did not have much variance. In this particular example the model learned to apply the optimal action $[1\ 0\ 0]$ after observing middle and low CR values, but not after high CR values. In Panel B the Q-function values of an example model trained on RW patients with an fixed buffer that used the random extinction strategy and contained random actions are shown. The random buffer contained all possible actions with approximately the same frequency. It can be seen that in the random buffer condition the Q-values for all actions had similar variance over the learning steps. Interestingly, in this particular example the model did not find the optimal action for the three observed CR levels. While after observing high and middle CR levels the model performed action $[1\ 1\ 0]$, after observing low CR levels it even performed $[1\ 0\ 1]$, and therefore it reinforced the previously learned association between cue A and the US. In Panel C the Q-function values for a model which used an empty non-fixed buffer trained on RW patients can be seen. The model tended to generate the optimal action $[1\ 0\ 0]$ after observing high, middle, and low CR levels. In summary no benefits in using the fixed version of the buffer were observed for training with simulated data, learning might have been impaired depending on the experiences in the buffer, and models that were trained on buffers containing random actions resulted on average in higher rewards, but when examined more closely performed more poorly on low CRs.

A



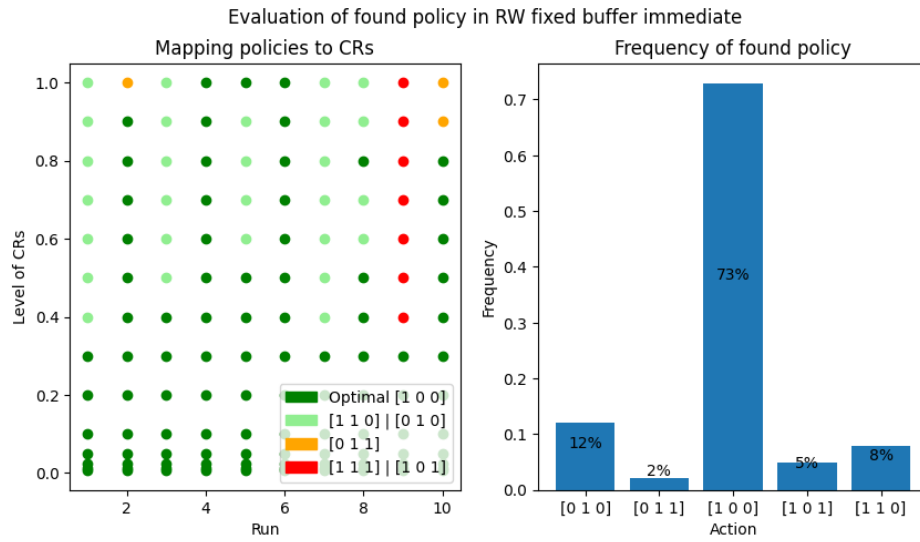
B

**Figure A1**

Training curves for DQN trained on RW patients – comparing fixed and non-fixed buffers

Note. Panel A: Comparison of achieved mean rewards during training of DQN therapists on RW patients. The fixed buffer was adapted to not replace older with newer experiences during training. Panel B: Comparison of training curves of DQN trained on RW patients with random actions filled in the buffer. IQR, the inter-quartile range, is shown at the lighter colored areas. No apparent performance differences between fixed and non-fixed buffers.

A



B

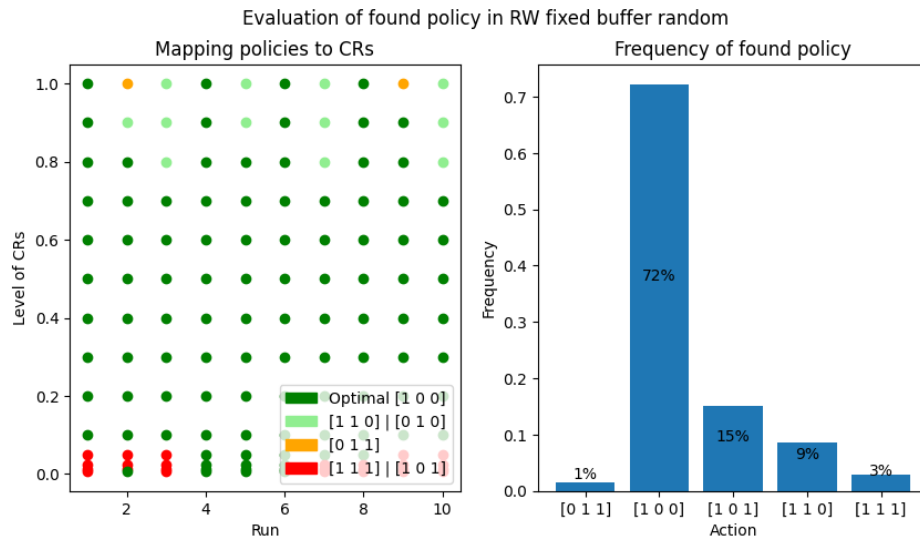
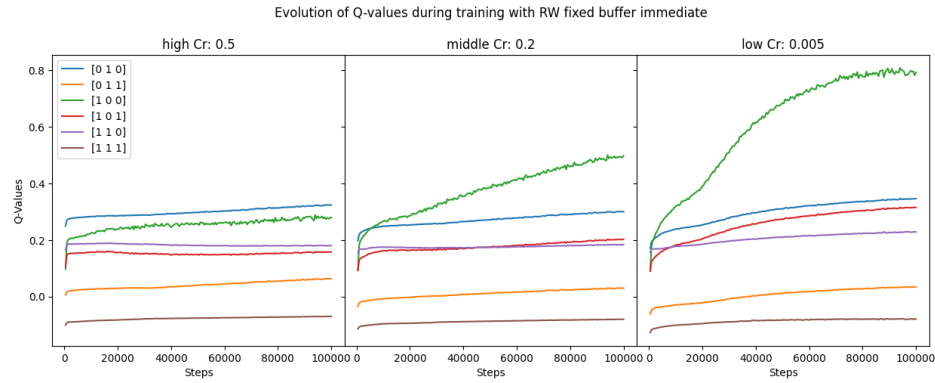


Figure A2

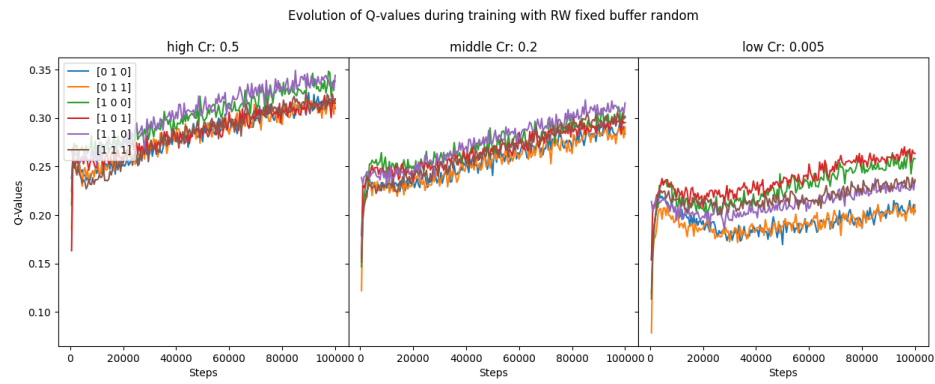
Policy evaluation of DQN with a fixed buffer trained on RW patients

Note. Panel A: Evaluation of learned policies during training with an immediate fixed buffer. One trained RL therapist appears to be an outlier with frequent action [1 0 1]. Panel B: Evaluation of learned policies during training with a random fixed buffer. RL therapists learned to perform [1 0 1] or [1 1 1] for low CRs.

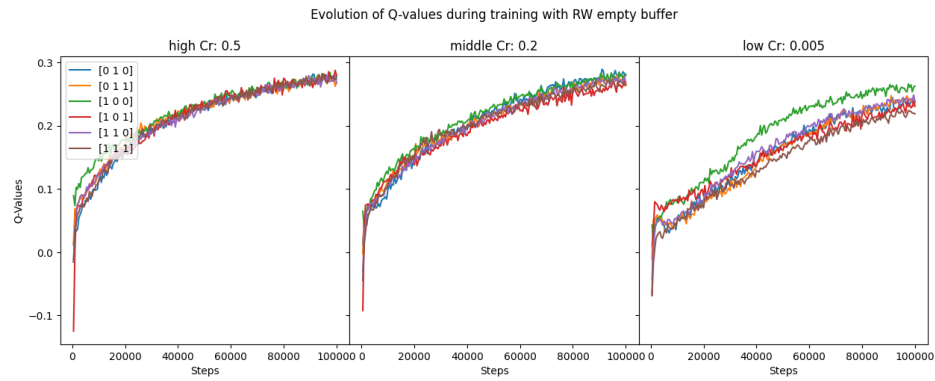
A



B



C

**Figure A3***Evolution of Q-values during training of DQN*

Note. The plot shows the evolution of the DQN's Q-value function. Plotted are the Q-values of all six available actions, when observing three different CRs. Panel A: Evolution of Q-values for DQN trained on RW patients using a fixed buffer filled with the action [1 0 0] only. Panel B: Evolution of Q-values for DQN trained on RW patients using a fixed buffer filled with random actions. Panel C: Evolution of Q-values for DQN trained on RW patients with an empty non-fixed buffer.

Experiment A2

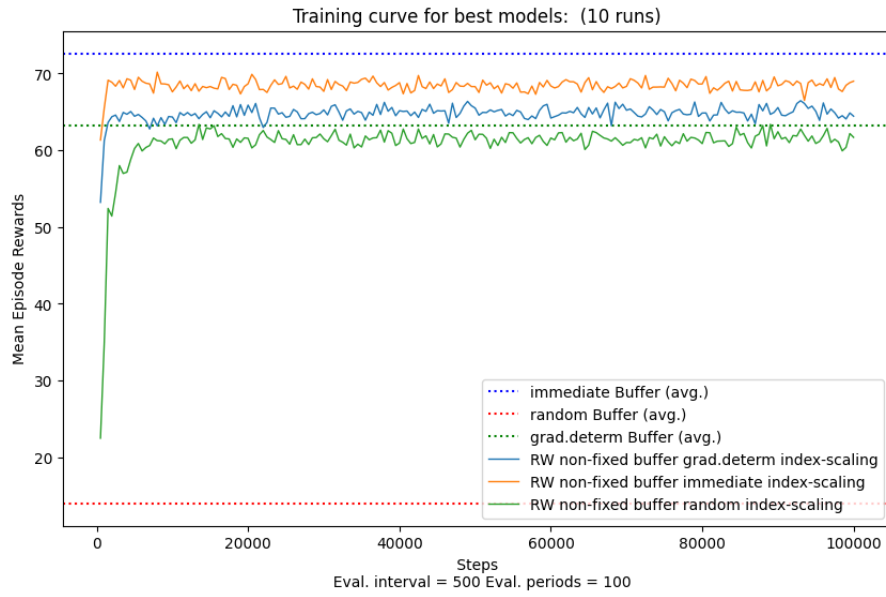
The goal of experiment [A2](#) was to evaluate, whether adding a time dimension to the observed data could facilitate or improve the RL therapist’s performance on RW patients. When applying RL to healthcare decision making it is desired to provide the algorithm with at least the same amount of variables that would be considered by a human clinician (Gottesman et al., [2018](#), [2019](#)). So far, the RL therapists have only considered one variable, the observed CR, for their decision which stimuli to show. Other potentially relevant variables were omitted for simplicity. For instance, a therapist would consider the duration and progress of therapy for subsequent decisions. Thus, in Experiment [A2](#) it was evaluated what impact a second dimension that contained a measure of time or progress had on the performance and training of the RL therapist. Thus, to inform the therapist about the progress index-scaling was applied. The therapy in stage 2 consisted of 24 trials. The index of the current trial was transformed to a normalized time point ($\in [0, 1]$), computed as the current trial number divided by 24. The index-scaled time point was passed to the therapist alongside the CR to indicate therapy progress.

In Figure [A4](#) Panel A the training curves for models with a default buffer that were non-fixed and that contained experiences from an immediate, gradually deterministic, or a random extinction strategy, and that were trained on RW patients are shown. The other extinction conditions were omitted to save computational capacity and were not expected to have superior performances or to contribute to evaluating whether an additional time dimension would be beneficial for training or performance. In Panel A the model trained with a buffer filled with immediate extinction achieved on average the highest rewards. However, it did not reach the levels as high as the baseline comparison with the manually applied immediate extinction strategy. In addition to applying index-scaling, in Panel B the fixed version of the buffer was added. Likewise, the level of the baseline comparison was not reached. Among the models trained with a fixed buffer and index-scaling the training models with gradually deterministic experiences performed best. Surprisingly,

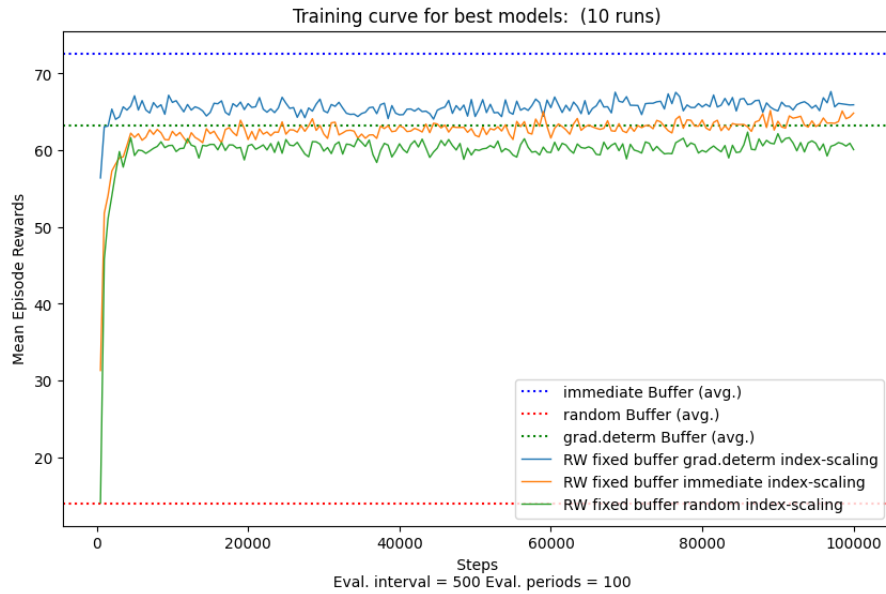
when applying index-scaling, for both the models trained with and without a buffer adaptation the models using random actions in the buffer achieved on average the lowest rewards on average, as seen in Panel A and B. In Figure [A5](#) it is shown that training with the same extinction strategy and the same non-fixed buffer achieved consistently higher rewards with a one dimensional observation space and without index-scaling compared to the two dimensional index-scaling condition. In Figure [A6](#) it is shown that increasing the training steps to 500000 yielded average rewards as high as the baseline comparison, when applying index-scaling with an fixed buffer and immediate extinction experiences but not with the original buffer version. Furthermore, in Figure [A6](#) the IQR of the training curves was plotted, and it suggested that 25% of the reward values came close to the baseline comparison level.

In Figure [A7](#) the learned policies of the best models trained when index-scaling was applied were evaluated. In Panel A and B the first and second training runs that used a fixed buffer are illustrated. In Panel C and D the first and second training runs that used the original non-fixed buffer are illustrated. For simplicity and space the other runs were not plotted. The X-axis shows the index-scaled time point and the Y-axis shows the corresponding observed CR value. The color of the plotted dots indicates which policy was used and in the legends in can be seen which color corresponded to which action. Overall, the plots show that the models learned to perform the optimal policy $[1 \ 0 \ 0]$ often. However, sometimes they still found a policy which was sub-optimal, or even reinforced the association between cue A and the US. While it might be desired to include as many variables during training as would be considered by clinicians for their decision making, there were no observed benefits in regard to finding the optimal behavior sooner or more often. On the contrary, it added complexity to the simulation, and thus the RL therapists required more training steps to learn to perform ET or a meaningful policy.

A



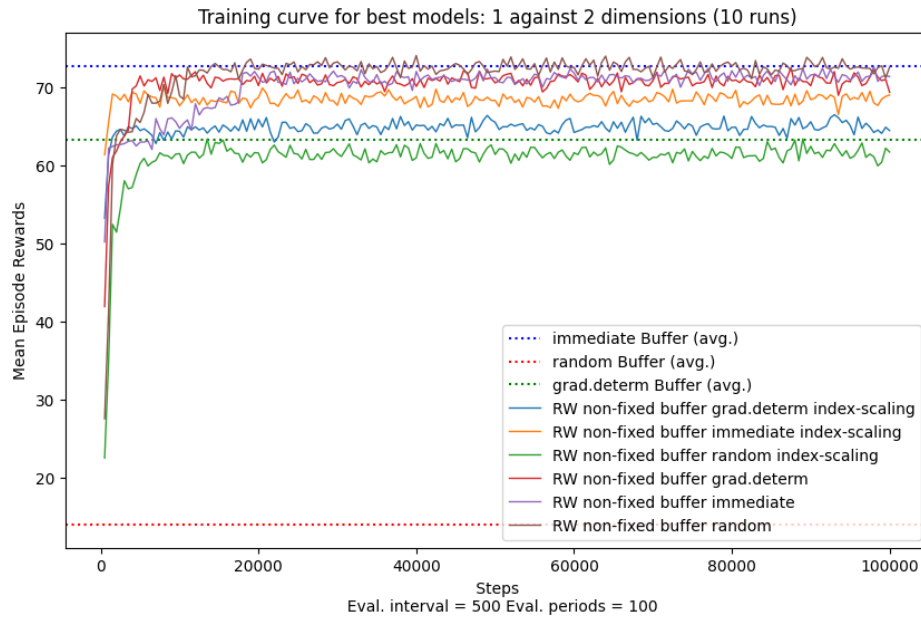
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**Figure A4**

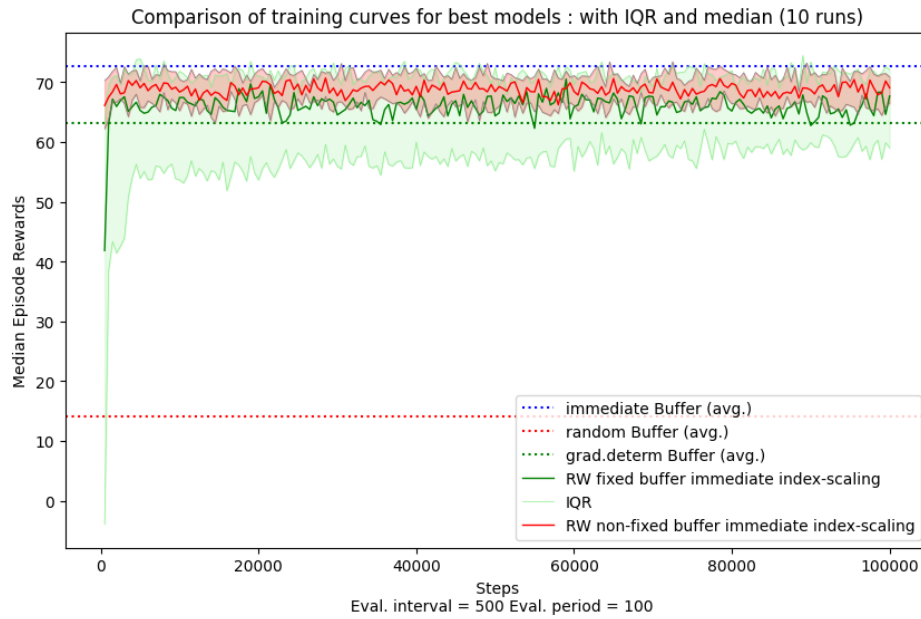
Training curves for DQN trained on RW patients with a two dimensional (index-scaling) observation space

Note. The second dimension used index-scaling, and was calculated by $\frac{n_{trial}}{24}$. It provides the RL therapist with a temporal reference point during ET. Panel A: DQN trained on RW patients with a non-fixed buffer and index-scaling. Panel B: DQN trained on RW patients with a fixed buffer and index-scaling.

A

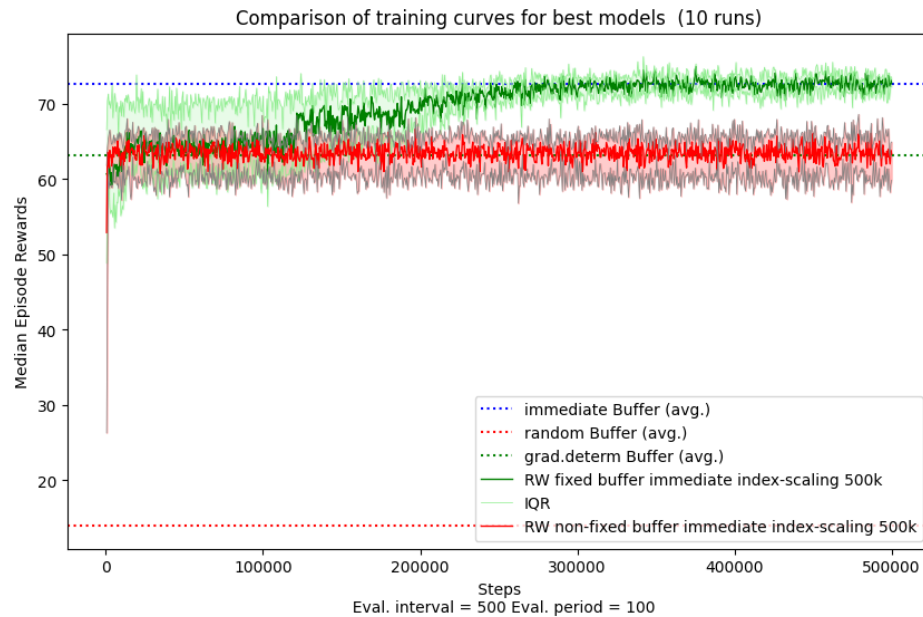


B

**Figure A5**

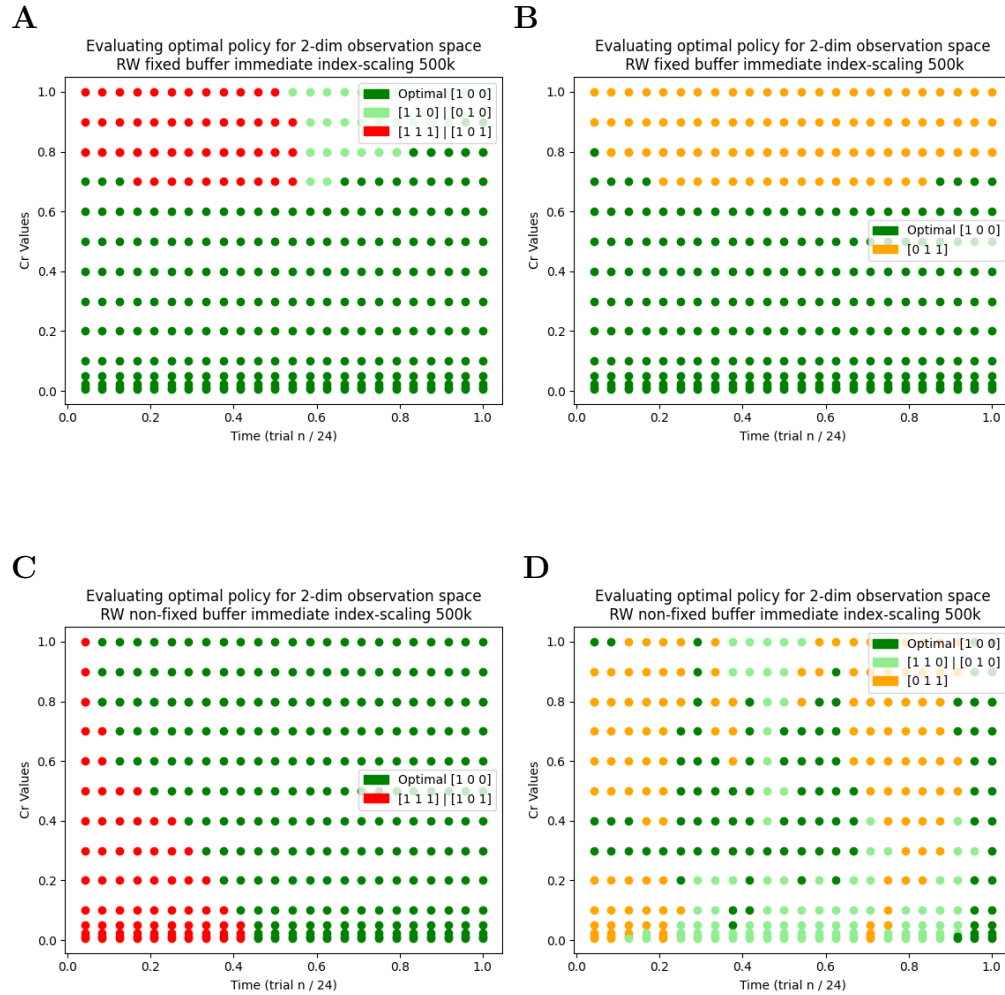
Training curves for DQN trained on RW patients with a one and two dimensional observation space

Note. The second dimension used index-scaling, and was calculated by $\frac{n_{trial}}{24}$. It provides the RL therapist with a temporal reference point during ET. Panel A: Comparison 1 against 2 dimensional observation space. Panel B: Comparison fixed and non-fixed immediate buffer with index-scaling.

**Figure A6**

Training curves for DQN trained on RW patients with a two dimensional observation space

Note. Both training curves shown used index-scaling, in addition to the observed CR level, during training. The green line was trained with a fixed buffer, and the red line without adaptation. The training steps were increased to 500000. See Experiment [A2](#)

**Figure A7**

Evaluation of learned policy when training DQN with two dimensional observation space

Note. Panel A: Run 1 as an example of DQN trained on RW patients with a fixed buffer filled with [1 0 0] action only. Panel B: Run 2 as an example of DQN trained on RW patients with a fixed buffer filled with [1 0 0] action only. Panel C: Run 1 as an example of DQN trained on RW patients with a non-fixed buffer filled with [1 0 0] action only. Panel D: Run 2 as N example of DQN trained on RW patients with a non-fixed buffer filled with [1 0 0] action only.